Computers & Industrial Engineering 74 (2014) 270–290

Contents lists available at ScienceDirect

Computers & Industrial Engineering

journal homepage: www.elsevier.com/locate/caie

An artificial bee colony algorithm approach for the team orienteering problem with time windows ☆

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ARTICLE INFO

Article history:
Received 30 July 2012
Received in revised form 18 January 2014
Accepted 5 June 2014
Available online 16 June 2014

Keywords:
Heuristics
Routing
Team orienteering problem
Time window
Artificial bee colony algorithm

ABSTRACT

This study addresses a highly constrained NP-hard problem called the team orienteering problem with time windows (TOPTW), which belongs to a well-known class of vehicle routing problems. This study proposes a relatively new technique called artificial bee colony (ABC) approach to solve the TOPTW. Moreover, considering that the number of studies for discrete optimization with an ABC algorithm is comparatively low, this study presents a new use of the ABC algorithm for a difficult discrete optimization problem. Additionally, this study introduces a new food source acceptance criterion and a new scout bee search behavior, both of which significantly contribute to the solution quality. The results show that the proposed method is effective, efficient, and comparable to other approaches.

1. Introduction

The team orienteering problem with time windows (TOPTW) is a problem that has been widely studied by researchers and practitioners. In fact, the TOPTW belongs to the class of well-known vehicle routing problems. To describe that class of problems, we can briefly state that a fleet of vehicles must serve all of a given set of customers with respect to some constraints. However, few problems consider assigning a profit to each customer. When the customers are associated with profits, serving a subset of the customers with respect to given constraints is allowed to maximize the total profit. When the TOPTW is considered, the objective is to collect maximum profits by visiting customers within given time windows. That is, each customer can be visited at most once, and the visit must start within the customer's time window. Although some researchers have focused on the case in which a single vehicle is available (Feillet et al., 2005), the TOPTW considers multiple vehicles to serve a set of profitable customers. However, the vehicle fleet does not have to serve the entire customer set. Instead, it is possible that only a customer subset is to be visited subject to time window constraints.

We must note that the TOPTW is an extension of the team orienteering problem (TOP), in which a set of vehicle tours are constructed such that the total collected reward received from visiting a subset of customers is maximized and the length of each vehicle tour is restricted by a prespecified limit. The TOP is also an extension of the orienteering problem (OP) (Chao, Golden, & Wasil, 1996; Tsiligirides, 1984). Interested readers can find detailed research in the studies of Vansteenwegen, Souffriau, and Van Oudheusden (2011) and Tang and Miller-Hooks (2005) for the TOP and OP.

Because the TOPTW is a highly constrained NP-hard combinatorial optimization problem, exact solution approaches are incapable of solving large-scale TOPTW instances within a reasonable computational time (Kantor & Rosenwein, 1992). Thus, almost all of the prior studies cover heuristic or meta-heuristic approaches. However, to our knowledge, none of these studies uses an artificial bee colony (ABC) approach, which is the most recently introduced heuristic technique (Karaboga, 2005). Although Szeto, Yongzhong, and Ho (2011) introduced an ABC heuristic for the capacitated vehicle routing problem, they did not focus on the TOPTW. Additionally, although there are many approaches for solving the TOP in the literature (Archetti, Hertz, & Speranza, 2007; Bouly H. & Moukrim. A., 2010; Butt & Cavalier, 1994; Dang, Guibadj, & Moukrim, 2013; Ke, Archetti, & Feng, 2008; Souffriau, Vansteenwegen, Berghe, & Van Oudheusden, 2010; Tang & Miller-Hooks, 2005; Vansteenwegen, Souffriau, Berghe, & Van Oudheusden, 2009a), it is still possible that these approaches may not be efficient or may be ineffective when attempting to solve the TOPTW. Some of the studies that are related to team orienteering cover a Traveling Salesman Problem (TSP) heuristic aiding an exact enumerative algorithm (Laporte & Martello, 1990), a greedy heuristic algorithm (Butt & Cavalier, 1994), a local search method (Chao et al., 1996; Vansteenwegen, Souffriau,

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http://dx.doi.org/10.1016/j.cie.2014.06.004
0360-8352/

Berghe, & Van Oudheusden, 2009b; Vansteenwegen et al., 2009a), tabu searching (Tang & Miller-Hooks, 2005), an Ant Colony System (ACS) approach (Montemanni & Gambardella, 2009), Variable Neighborhood Search algorithms (Labadie, Mansini, Melechovský, & Calvo, 2012; Tricoire, Romauch, Doerner, & Hartl, 2010), a simulated annealing (SA) approach (Lin & Yu, 2012), a SA and Local Search combined algorithm (Hu & Lim, 2014), and a hybridized evolutionary local search algorithm (Labadie, Melechovský, & Calvo, 2011).

Laporte and Martello (1990) studied a problem that is referred to as the Selective Travelling Salesman Problem (STSP). The STSP corresponds exactly to the OP. The investigators have suggested a TSP heuristic aiding exact enumerative algorithms while embedding upper and lower bounds.

Butt and Cavalier (1994) studied the Multiple Tour Maximum Collection Problem, which is in the class of multi-vehicle routing problems. Practical application of this problem was in athlete recruiting from high schools. Although there is no vehicle, each day's tour corresponds to a vehicle route. And an athlete recruiter visits chosen high schools to maximize the recruitment potential of his/her college in a tour. The authors proposed a greedy heuristic algorithm to solve this problem.

Chao et al. (1996) suggested a local search method to solve the TOP, in which a set of vehicle tours are constructed such that the total collected reward received from visiting a subset of customers is maximized, and the length of each vehicle tour is restricted by a prespecified limit. Additionally, Tang and Miller-Hooks (2005) exploited the tabu search heuristic, and Vansteenwegen et al. (2009a) used a guided local search to solve the TOP.

Montemanni and Gambardella (2009) introduced an efficient ACS algorithm that focuses on solving the TOPTW. Their algorithm is based on the solution of a hierarchical generalization of the TOPTW. Another efficient algorithm for solving the TOPTW was the iterated local search algorithm of Vansteenwegen et al. (2009b). Their algorithm combines an insert move with a shake step to escape from a local optimum.

Lastly, state-of-the-art studies are presented by Tricoire et al. (2010), Labadie et al. (2011), Lin and Yu (2012), and Labadie et al. (2012), which address the TOPTW. Interestingly, the Variable Neighborhood Search (VNS) heuristic of Tricoire et al. (2010) was originally designed for the multi-period orienteering problem with multiple time windows. However, this algorithm finds good solutions for the TOPTW as well. Labadie et al. (2011) presented an effective hybrid metaheuristic to solve both the OP and the TOP with time windows. Their method combines the greedy randomized adaptive search procedure (GRASP) with the evolutionary local search (ELS). The ELS generates multiple distinct child solutions using a mutation mechanism. Each child solution is further improved by a local search procedure. The GRASP provides multiple starting solutions to the ELS. Lin and Yu (2012) used three neighborhood search operators in their SA algorithm, which are swap, insertion, and inversion moves. After each move, a new solution is accepted if the new objective function value is better. Otherwise, it is accepted with a small probability. Labadie et al. (2012) proposed a Granular Variable Neighborhood Search (GVNS) procedure based on the idea of exploring, most of the time, granular rather than complete neighborhoods to improve the algorithm's efficiency without losing effectiveness. Hu and Lim (2014) proposed an iterative framework incorporating three components to solve the TOPTW. The first two components are a local search procedure and a simulated annealing procedure. They explore the solution space and discover a set of routes. The third component recombines the routes to identify high quality solutions.

This study focuses on solving the TOPTW, which belongs to the class of vehicle routing problems; we use a relatively recently introduced heuristic method, the ABC algorithm based on the foraging behavior of honey-bees, which is typically used for numerical optimization problems. The ABC algorithm was initially proposed for numerical function optimization (Karaboga & Basturk, 2007) and, to our knowledge, was first used for discrete optimization problems by Singh (2009). The number of studies that exploit the ABC algorithm for discrete optimization problems is still comparatively low. This study introduces an ABC algorithm for a difficult discrete optimization problem, the TOPTW, that is effective, efficient, and comparable to other methods.

The remainder of this study is organized as follows: Section 2 defines the TOPTW, and Section 3 briefly defines the ABC technique. Section 4 presents the proposed method. Section 5 provides computational experiments. Lastly, Section 6 concludes the paper.

2. The team orienteering problem with time windows

Based on the notation of Lin and Yu (2012), the TOPTW can be defined on a graph $G = (N, A)$, where $N = \{1, \ldots, n\}$ is the set of locations and $A = \{(i, j): i \neq j \wedge i, j \in N\}$ is the set of arcs that connect the locations. Each location $i = \{1, \ldots, n\}$ is associated with a profit S_i, a service time T_i, and a time window $[O_i, C_i]$. The number of tours must be equal to m. Each tour starts from location 1 and ends in location n. The travel time from location i to location j is denoted by t_{ij}. The total travel time of a route cannot exceed the given time limit $T_{\max}$. Each visit must start within the location's time window. However, if a vehicle arrives at a location before its time window starts, then that vehicle is allowed to wait at that location. Each location's profit can be collected only by one vehicle. The goal of the TOPTW is to find a route for each tour such that the total profit is maximized. Thus, the TOPTW can be stated as follows (Vansteenwegen et al., 2009a):

$$\text{maximize} \sum_{d=1}^{m}\sum_{i=2}^{n-1} s_i y_{id} \tag{1}$$

$$\text{subject to} \sum_{d=1}^{m}\sum_{j=2}^{n-1} x_{1jd} = \sum_{d=1}^{m}\sum_{i=2}^{n-1} x_{ind} = m, \tag{2}$$

$$\sum_{d=1}^{m} y_{dk} \le 1, \quad k = 2, \ldots, n-1, \tag{3}$$

$$\sum_{i=1}^{n-1} x_{ikd} = \sum_{j=2}^{n-1} x_{kjd} = y_{kd}, \quad k = 2, \ldots, n-1,$$
$$d = 1, \ldots, m, \tag{4}$$

$$\sum_{i=1}^{n-1}\left(T_i y_{id} + \sum_{j=2}^{n} t_{ij} x_{ijd}\right) \le T_{\max}, \quad d = 1, \ldots, m, \tag{5}$$

$$\alpha_{1d} = T_1 \quad d = 1, \ldots, m, \tag{6}$$

$$\alpha_{rd} = \max\left[\left(\alpha_{(r-1)d} + \sum_{i=2}^{n}\sum_{j=1}^{n-1}\begin{cases} t_{ij}x_{jid}y_{id} & \text{if } u_{id} = r \\ 0 & \text{else}\end{cases}\right),\right.$$
$$\left.\sum_{i=1}^{n}\begin{cases} O_i y_{id} & \text{if } u_{id} = r \\ 0 & \text{else}\end{cases}\right] + \sum_{i=1}^{n}\begin{cases} T_i y_{id} & \text{if } u_{id} = r \\ 0 & \text{else}\end{cases},$$
$$r = 2, \ldots n, d = 1, \ldots, m, \tag{7}$$

$$C_i \ge \sum_{d=1}^{m}\sum_{r=2}^{n}\begin{cases} y_{id}(\alpha_{rd} - T_i) & \text{if } u_{id} = r \\ 0 & \text{else}\end{cases}, \quad i = 1, \ldots, n, \tag{8}$$

$$2 \le u_{id} \le n, \quad i = 2, \ldots, n, d = 1, \ldots, m, \tag{9}$$

$$u_{id} - u_{jd} + 1 \le (n-1)(1 - x_{ijd}), \quad i = 2, \ldots, n,$$
$$d = 1, \ldots, m, \tag{10}$$

$$x_{ijd}, y_{id} \in \{0, 1\}, \quad i = 1, \ldots, n, d = 1, \ldots, m. \tag{11}$$

where y_{id} is 1 if location i is visited in tour d; otherwise, y_{id} is 0; x_{ijd} denotes that location j is visited immediately after location i in tour

d if $x_{ijd} = 1$, otherwise $x_{ijd} = 0$; u_{id} denotes the position of location i in tour d; and α_{rd} is the departure time from the location, which is at position r of tour d.

Eq. (2) guarantees that each tour starts from location 1 and ends at location n. Eq. (3) ensures that every location is visited at most once. Eq. (4) guarantees that if a location is visited in a given tour, it is preceded and followed by exactly one other visit in the same tour. Eq. (5) guarantees that each tour is completed within a given time limit. Eq. (6) calculates the departure times from the first locations in all of the tours. Eq. (7) calculates the departure time from the location that is at position r of tour d. Eq. (8) guarantees that each visit starts within a location's time window. Lastly, Eqs. (9) and (10) are necessary to prevent subtours.

3. Artificial bee colony algorithm

The ABC heuristic method simulates, in essence, the foraging behavior of honey-bee swarms. Honey-bees are among the most closely studied social insects. Their foraging behavior, learning, memorizing, and information-sharing characteristics have recently been one of the most interesting research areas in swarm intelligence (Teodorovic, Lucic, Markovic, & Orco, 2006). In the ABC algorithm of Karaboga and Basturk (2007), the colony of artificial bees contains three groups of bees: employed bees, onlookers, and scouts. A bee waiting in the dance area to make a decision to choose a food source that corresponds to a solution is referred to as an onlooker, and a bee going to the food source that it visited previously is referred to as an employed bee. A bee conducting a random search is referred to as a scout. Thus, the main steps of the algorithm are given as follows:

(1) Initialize.
(2) REPEAT.
  (a) Place the employed bees on the food sources in the memory;
  (b) Place the onlooker bees on the food sources in the memory;
  (c) Send the scouts to the search area for discovering new food sources.
(3) UNTIL the stopping condition is met.

At the initialization stage, the food sources are chosen randomly. Then, at each iteration, each employed bee moves between two randomly chosen food sources. The employed bee measures the amount of nectar, which actually is the objective function value, at the new food source. Each onlooker bee moves between two food sources that are chosen randomly with respect to a probability. That is, an onlooker prefers a food source depending on the nectar information. Thus, as the nectar amount of a food source increases, the probability with which that food source is chosen by an onlooker increases also. When a food source is exhausted by an employed or onlooker bee, it is randomly generated by scout bees, as if it was initialized.

4. Proposed artificial bee colony approach

As clearly shown in the previous section, artificial bees continuously move to search for food sources with a larger amount of nectar. In this study, each food source denotes a produced set of routes, and the amount of nectar at a given food source as shown in Eq. (1). To escape from a local optimum in the original ABC method, if a food source position cannot be improved further through a predetermined number of iterations, then that food source is assumed to be abandoned (Karaboga & Basturk, 2007). Thus, that food source is reinitialized. However, to solve the TOPTW effectively, we suggest using scout bees to apply a local search over all of the possible neighbors at random iterations with a small probability.

4.1. Solution representation

Before introducing the solution representation, we must note that there is an important difference between the mathematical programming model described in Eqs. (1)–(11), which basically depends on the model provided by Vansteenwegen et al. (2009a), and the benchmark instances. According to the mathematical model, each vehicle starts its tour from location 1 and ends at location n. However, in the benchmark problem sets, a TOPTW tour is converted into a tour that starts from the depot and ends at the depot. Thus, the maximum time limit of a tour is turned into the latest starting time of the depot's time window.

After that explanation, similarly to the solution representation of many researchers who focused on the class of well-known vehicle routing problems, we represent a TOPTW solution by a string of numbers that consists of a permutation of n locations, which is denoted by the set $\{1, 2, \ldots, n\}$ and $m - 1$ negative numbers $\{-1, -2, \ldots, 1 - m\}$ to separate the tours. That is, each substring between the negative numbers corresponds to a tour, and the jth nonnegative number in a substring indicates the jth location to be visited in that tour. Thus, the first nonnegative number in a solution indicates the first location to be visited in the first tour. Other locations are added to the tour one by one from left to right. However, as suggested by Lin and Yu (2012), if adding a location to the current tour violates a location's time window constraint, then that location is skipped. Therefore, the involved location becomes an unrouted location. If a tour exceeds the maximum duration, then all locations in that tour become unrouted locations, and that tour collects no profit. For a problem with 20 locations and 5 tours, a sample solution is shown in Fig. 1. Note that the italicized numbers indicate the unrouted locations. As mentioned previously, a solution corresponds to a food source in our ABC heuristic. Thus, at this sample food source, the first tour starts from the depot; then, it visits locations 4, 6, and 3 successively and finally returns to the depot. The second tour visits locations 5, 20, and 1; the third tour visits locations 2, 12, 16, 8, and 13; the fourth tour visits locations 10 and 14; and the final tour visits locations 15 and 11 (all visits are given in successive order).

According to the original ABC method, bees are initially sent to randomly generated food sources. Thus, our ABC algorithm starts searching with L randomly generated food sources at the initialization phase. The amount of nectar nc_l at each food source fs_l ($l = 1, \ldots, L$) is calculated. Additionally, the number of employed bees eb and the number of onlooker bees ob must be determined at this stage. The original ABC method assumes that the number of scout bees sb is zero at the initialization phase because a food source is abandoned by an employed bee if the quality of the food source has not been improved for a predetermined number of successive iterations; at this time, the employed bee becomes a scout. After the scout finds a new food source, it becomes an employed bee again. However, as mentioned previously, our ABC heuristic does not monitor the number of unimproved successive iterations. Thus, our ABC algorithm determines the number of scouts at the initialization phase as well.

4.2. Moving an artificial bee

Although there are three groups of bees, this study proposes two major moving strategies because employed and onlooker bees move in almost the same manner. As mentioned previously, a bee continuously moves between two food sources. Thus, only the food source selection mechanism differs between the moving strategies

4	6	*19*	3	-2	5	9	20	1	-4	2	12	16	8	*17*	7	13	-3	10	14	-1	15	*18*	11

Fig. 1. A sample food source.

of an employed bee and an onlooker bee. However, a scout bee moves completely differently; this will be discussed in detail later.

Let fs_l and fs_k denote two food sources around which a bee moves. The original moving operation was defined for a numerical function optimization problem by Karaboga and Basturk (2007) as follows:

$$x_i(fs_l) = x_i(fs_l) + \varphi(x_i(fs_l) - x_i(fs_k)) \tag{12}$$

where $x_i(fs_l)$ and $x_i(fs_k)$ denote the food source positions on the ith dimension and φ is a random number between $[-1, 1]$ that controls the production of a new food source position around $x_i(fs_l)$. However, Eq. (12) is not applicable for most of the discrete optimization problems such as the TOPTW. For that reason, when solving a capacitated vehicle routing problem, Szeto et al. (2011) suggested that a bee chooses only one food source fs_l and moves around it with seven neighborhood operators, namely, swap, swap of subsequences, insertion, insertion of subsequences, reversing of a subsequence, swap of reversed subsequences, and insertion of reversed subsequences. However, the original ABC method is based on information sharing among the food sources, as can clearly be seen in Eq. (12).

Thus, we suggest that a bee randomly chooses two food sources fs_l and fs_k. Then, it randomly selects a location i (including negative tour separators) and finds position r_i of that location in both of these food sources. We then suggest changing the position of location i from $r_i(fs_l)$ to $r_i(fs_k)$ in food source fs_l. Thus, the position of location i would be the same in both of the food sources. To change the position of location i from $r_i(fs_l)$ to $r_i(fs_k)$, our ABC algorithm exploits three move operators, which are swap, insertion and inversion moves. In fact, these move operators are very common among the solution methods for vehicle routing class problems (Lin & Yu, 2012; Szeto et al., 2011; Tricoire et al., 2010).

The swap move selects the $r_i(fs_l)$th and the $r_i(fs_k)$th locations of fs_l and then exchanges their positions. See Fig. 2, where $i = 5$, $m = 3$, and $n = 8$. The insertion move is performed by selecting the $r_i(fs_l)$th location of fs_l and then inserting it into the position immediately after the $r_i(fs_k)$th location. An example is shown in Fig. 3. The inversion move is performed by reversing the sequence between two locations (including two selected locations), as shown in Fig. 4. According to our method, when an artificial bee is searching around a food source, it randomly performs one of these move operators with equal probability.

According to the above explanation, one can infer that food source fs_k does not play a role when a bee is moving and food sources do not really share information. However, especially at the higher iterations, the number of similar food sources would be increased. Thus, if the position of location i is the same in many food sources at a given iteration, our movement method would preserve the location of i with a high probability.

The new food source acceptance criterion of the original ABC method is very strict. That is, the new food source fs_{new} is accepted only if the nectar amount of fs_{new} is higher than that of fs_l; otherwise, it is rejected. However, a scout does not reinitialize a food source when it is stuck in a local optimum in our ABC heuristic; to escape from a local optimum, we suggest accepting a new food source with a small probability even if the new food source is worse. Let $\Delta = obj(fs_{new}) - obj(fs_l)$. In our method, the probability of replacing fs_{new} with fs_l is $e^{\Delta/\rho}$. This probability calculation is quite similar to the probability calculation of an increase in energy of a SA heuristic at temperature ρ. In the SA method, the temperature is reduced iteratively with respect to a cooling schedule. However, ρ in [0, 1] is a constant parameter in our algorithm, and provided that Δ is greater than or equal to zero, a new food source would always be replaced with the current source according to this calculation.

				$r_i(fs_l)^{th}$ location ↓				$r_i(fs_k)^{th}$ location ↓		
fs_l (before):	5	4	-2	6	2	-1	1	8	7	3
fs_{new} (after):	5	4	-2	8	2	-1	1	6	7	3

Fig. 2. Swap move.

				$r_i(fs_l)^{th}$ location ↓				$r_i(fs_k)^{th}$ location ↓		
fs_l (before):	5	4	-2	6	2	-1	1	8	7	3
fs_{new} (after):	5	4	-2	2	-1	1	8	6	7	3

Fig. 3. Insertion move.

				$r_i(fs_l)^{th}$ location ↓				$r_i(fs_k)^{th}$ location ↓		
fs_l (before):	5	4	-2	6	2	-1	1	8	7	3
fs_{new} (after):	5	4	-2	8	1	-1	2	6	7	3

Fig. 4. Inversion move.

As previously mentioned, employed and onlooker bees move in the same manner; the only difference between them is the selection of food sources around which they move. An employed bee chooses food sources randomly with equal probability. However, an onlooker bee chooses a food source depending on the probability value associated with that food source, $prob_l$, which is calculated as follows:

$$prob_l = \frac{nc_l}{\sum_{k=1}^{L} nc_k} \tag{13}$$

This slight difference between the moving strategies of onlooker and employed bees causes the food sources to not converge rapidly. Because an onlooker bee gives more chances to food sources with higher amount of nectar, the probability of moving to alike food sources rises. Thus, number of alike food sources would be iteratively increased if only onlooker bees are used. However, employed bees give equal chances to all food sources, which would diversify the food sources at a higher number of iterations. In contrast, moving around food sources with a higher amount of nectar would increase the probability of finding a new food source that also has a high amount of nectar. Therefore, our ABC algorithm includes both onlooker and employed bees.

The movement procedure is shown in Fig. 5. In the original ABC method, if a food source gets stuck in a local optimum for a predetermined number of iterations, then that food source is assumed to be abandoned (Karaboga & Basturk, 2007). However, because our solution acceptance criterion enables our algorithm to escape from local optima, we suggest that a scout bee uses a local search procedure that first applies all of the possible swap moves to a randomly chosen food source fs_l. Then, all of the possible insertion moves are applied to fs_l. The reason why a scout bee does not

```
Set i = a random location (including negative tour separators)
Set l = randomly select a food source (with equal probability for an employer bee or with
    respect to Eq. (13) for an onlooker bee)
Set k = randomly select a food source (with equal probability for an employer bee or with
    respect to Eq. (13) for an onlooker bee)
Set r_i(fs_l) = position of i in food source fs_l
Set r_i(fs_k) = position of i in food source fs_k
Set rand = a random number between 0 and 1
if rand < 1/3 then
   Set fs_new = swap the r_i(fs_l)th location with the r_i(fs_k)th location of fs_l   // see Fig. 2.
else if rand < 2/3 then
   Set fs_new = insert the r_i(fs_l)th location into position r_i(fs_k) of fs_l     // see Fig. 3.
else
   Set fs_new = reverse the sequence between the r_i(fs_l)th location and the r_i(fs_k)th location
      of fs_l // see Fig. 4.
end if
Δ = obj(fs_new) − obj(fs_l)
If a random number between 0 and 1 ≤ e^(Δ/ρ) then
    copy fs_new to fs_l
end if
```

Fig. 5. The movement procedure for employed and onlooker bees.

```
set fs_l = randomly select a food source
for i = 0 to n + m − 1
    for j = 0 to n + m − 1
       set fs_new = swap the ith location with the jth location of fs_l   // see Fig. 2.
       if obj(fs_new) > obj(fs_l) then
          copy fs_new to fs_l
       end if
    end for
end for
for i = 0 to n + m − 1
    for j = 0 to n + m − 1
       set fs_new = insert the ith location into j position of fs_l   // see Fig. 3.
       if obj(fs_new) > obj(fs_l) then
          copy fs_new to fs_l
       end if
    end for
end for
```

Fig. 6. The movement procedure for scout bees.

exploit an inversion move operator is discussed in Section 5.2. The movement procedure for a scout bee is shown in Fig. 6.

Because a scout bee's search around a randomly chosen food source is very time consuming, we suggest that scout bees should not be sent to food sources at each one of the iterations. Instead, they should be sent to food sources at randomly chosen iterations with a very small probability μ.

At the end of each iteration, the best food source with the highest amount of nectar is compared with the best solution obtained thus far. If the best food source has a higher objective function value, then it is memorized as the best solution obtained thus far.

Consolidating all that we have discussed until now, the proposed ABC algorithm for the TOPTW is given in Fig. 7.

5. Computational experiments

In this section, we present the results that we obtained when searching for the solution of the TOPTW. The ABC approach of this study has been compared with seven other algorithms proposed by Hu and Lim (2014) (I3CH), Labadie et al. (2012) (GVNS), Lin and Yu (2012) (SSA), Labadie et al. (2011) (GRASP-ELS), Tricoire et al. (2010) (VNS), Vansteenwegen et al. (2009b) (ILS), and Montemanni and Gambardella (2009) (ACS), which are very effective and efficient meta-heuristics according to the results given by the authors. Although Lin and Yu (2012) introduced two versions of SA algorithms, we prefer comparing our approach with only SSA, which is more effective and can solve the TOPTW instances

```
Entirely Randomly initialize all of the food sources
Copy fs_best to X_best
while not (termination condition) do
   Send the employer bees to the food sources // see Fig. 5.
   Calculate the probabilities of the food sources // see Eq. (13)
   Send the onlooker bees to the food sources // see Fig. 5.
   If a random number between 0 and 1 < μ then
       Send the scouts to the food sources
   End if
   If obj(fs_best) > obj(X_best) then
        Copy fs_best to X_best
   End if
End while
```

Fig. 7. The proposed ABC approach for the TOPTW.

within a reasonable computational time. Because we could not obtain the original source codes of those algorithms, we prefer to present the original results of I3CH, GVNS, SSA, GRASP-ELS, VNS, ILS, and ACS, which are taken from the study of Labadie et al. (2011), Lin and Yu (2012), Labadie et al. (2012) and Hu and Lim (2014).

For evaluating the heuristics, we use the set of instances taken from Righini and Salani (2009), Montemanni and Gambardella (2009), and Vansteenwegen et al. (2009b), which are available on the Internet (Leuven, 2012) and include 75 TOPTW benchmark instances ranging from 48 to 288 possible visits. Each benchmark was run five times on an AMD Athlon X2 250 3.00 GHz computer running Microsoft Windows 7. The ABC algorithm was implemented in Java, and the Java codes were compiled under JDK 6.

The instances that were designed by Righini and Salani (2009) based on 48 instances in Solomon (1987) VRPTW data set and 10 instances in the multi-depot vehicle routing problems considered by Cordeau, Gendreau, and Laporte (1997). Montemanni and Gambardella (2009) developed another 37 instances, of which 27 instances are adapted from Solomon's (1987) data set and 10 instances from that of Cordeau et al. (1997). Vansteenwegen et al. (2009b) constructed a new data set of TOPTW instances which were extensions of those in Solomon (1987) and Cordeau et al. (1997). The major feature of this new data set is that the optimal solution for each instance is known. This is due to the specific setting on the number of provided vehicles, which enables all customers to be visited. Hence, the optimal objective value is equal to the total profit on the network graph (Hu & Lim, 2014).

The GVNS and GRASP-ELS algorithms were tested on the same desktop computer equipped with an Intel Pentium 4 3.00 GHz processor running under Windows XP Professional, which is comparable to our PC. The ACS and VNS heuristics were also run on a similar PC. However, ILS was run on a machine approximately two times faster than the computer on which GVNS and GRASP-ELS were tested (Labadie et al., 2012). The I3CH heuristic was tested on a Linux server equipped with an Intel Xeon E5430 2.66 GHz processor.

5.1. Parameter tuning

In our ABC heuristic, there are seven parameters that are used, namely, the *termination condition*, which denotes the maximum iteration number, the number of food sources L, the number of employed bees in the colony eb, the number of onlooker bees in the colony ob, the number of scout bees in the colony sb, the constant for the neighbor acceptance calculation ρ, and the probability of sending scout bees to food sources μ. To determine the appropriate parameter settings, we used a predetermined subset of 20 instances which we call "hard" instances. We firstly set initial parameter values as follows: *Termination condition* $= (n + m - 1) \times 5000$, $L = 10$, $eb = 10$, $ob = 5$, $sb = 1$, $\rho = 0.1$ and $\mu = 0.0001$. Then, we observed the performance of the ABC algorithm on each instance and select the ones that were performed worst. The chosen instances were r101, rc204, pr16 and pr20 with $m = 1$; r107, c204, pr06, pr14 and pr16 with $m = 2$; c203, c204, pr10, pr14 and pr16 with $m = 3$; r105, r110, pr06 and pr14 with $m = 4$; and c105 and c107 with $m = 10$. Table 1 gives the considered combinations of parameters that were tested on these 20 test instances and the selected values.

Table 1
Parameter combinations and selected values for the ABC heuristic.

Parameter	Considered values	Selected value
Termination condition	$(n + m - 1) \times 5000$, $(n + m - 1) \times 8000$, $(n + m - 1) \times 10{,}000$	$(n + m - 1) \times 8000$
L	10, 15, 20, 25, 30	10
eb	10, 15, 20, 25, 30	10
ob	5, 10, 15	5
sb	1, 2, 3, 4, 5, 10	1
ρ	0.1, 0.2, 0.3, 0.4, 0.5	0.3
μ	0.0001, 0.00015, 0.0002	0.00015

5.2. Comparison of move operators

To evaluate the effectiveness of each move operator, we tested our ABC heuristic on instances c101 with ten tours, r209 with three tours, pr01 with two tours, and pr04 with 12 tours by considering various combinations of move operators at a time, namely (1) swap move, (2) insertion move, (3) inversion move, (4) swap and insertion moves together, (5) insertion and inversion moves together, (6) swap and inversion moves together, and (7) all move operators combined. The results are reported in Table 2. The best-known solution for these instances are 1810, 1458, 502, and 2477, respectively. Each of the seven variants was run ten times, and each time the algorithm was run with the same parameter settings as given in Table 1, except for the number of scout bees, which was set to zero because scout bees apply all possible moves to a given food source. To observe the pure impact of a move operator on the solution quality, we prefer removing the scout bees from the colony for this subsection's analysis. For each one of the variants, Table 2 gives the maximum objective value that was obtained in ten runs, the minimum objective value in ten runs, the average objective value in ten runs, the standard deviation of the objective values in ten runs, the average percentage deviation from the best known solution, and the average CPU time in seconds for each run. As seen from the results, different move operators contribute differently. Interestingly, we noted that the inversion operator produces poor results, which is contrary to the results reported by Szeto et al. (2011). Additionally, Szeto et al. (2011) reported that the insertion operator obtained unsatisfactory solutions for the capacitated vehicle routing problem. However, the insertion operator obtains quite good solutions according to our test results. As far as the combined operators are concerned, we note that the inversion operator gives almost no contribution to the other operators. For that reason and for computational efficiency, scout bees do not exploit the inversion move operator in our ABC heuristic.

5.3. Impact of the acceptance criterion on the solution quality

According to the original ABC method, if a food source position cannot be improved further through a predetermined number of iterations, then that food source is assumed to be abandoned. When a food source is abandoned, some researchers prefer to randomly generate a new food source (Karaboga, 2005), whereas others generate a new food source by making minor random changes to the original source (Szeto et al., 2011). As previously mentioned, rather than counting nonimproving iterations, we suggest a flexible food source acceptance criterion. Thereby, scout bees can perform a wide local search of food sources at randomly chosen iterations with a small probability.

To compare our acceptance criterion with the original criterion, we exploit the original acceptance criterion according to which a new food source is accepted only if its amount of nectar is higher than that of the current source. Therefore, to escape from a local optimum, a scout bee must replace a food source fs that has been identified as not improving the solutions during the last *limit* successive iterations with a new solution. The quality of the new solution can be worse or better. The new solution is usually generated randomly. However, in the refined version of Szeto et al. (2011), a new solution is generated by applying a randomly chosen

Table 2
Experimental results using different operators.

Instance	Operator	Swap	Insertion	Inversion	Swap and insertion	Insertion and inversion	Swap and inversion	Combined
c101 (m = 10)	Maximum	1740	1720	1320	1780	1740	1730	1780
	Minimum	1680	1690	1250	1750	1680	1680	1730
	Average	1705	1709	1295	1764	1702	1705	1765
	Std. dev.	15.0	8.3	21.6	8.0	17.2	14.3	16.3
	Gap (%)	5.8	5.6	28.5	2.5	6.0	5.8	2.5
	Time (s)	38.5	32.6	28.7	22.9	33.6	44.6	27.0
r209 (m = 3)	Maximum	1415	1393	1078	1448	1419	1409	1449
	Minimum	1367	1369	1020	1423	1370	1368	1408
	Average	1387	1384	1057	1435	1388	1389	1437
	Std. dev.	12.2	6.7	17.6	6.5	14.0	11.6	13.3
	Gap (%)	4.9	5.1	27.5	1.6	4.8	4.7	1.5
	Time (s)	0.6	0.5	0.4	0.4	0.5	0.7	0.4
pr01 (m = 2)	Maximum	487	484	369	500	488	485	500
	Minimum	470	475	350	492	471	471	486
	Average	477	480	362	495	477	478	496
	Std. dev.	4.2	2.3	6.0	2.2	4.8	4.0	4.6
	Gap (%)	4.9	4.3	27.8	1.3	5.0	4.8	1.2
	Time (s)	2.6	2.2	2.0	1.6	2.3	3.1	1.9
pr04 (m = 12)	Maximum	2392	2383	1817	2461	2411	2390	2458
	Minimum	2310	2341	1721	2420	2328	2321	2389
	Average	2344	2367	1783	2439	2358	2356	2438
	Std. dev.	20.6	11.5	29.7	11.1	23.8	19.8	22.5
	Gap (%)	5.4	4.4	28.0	1.5	4.8	4.9	1.6
	Time (s)	172.4	147.1	128.7	103.1	151.6	200.7	121.4

move operator to *fs*. According to the results given by the authors (Szeto et al., 2011), their refined version of the ABC heuristic obtained better solutions. Thus, we preferred the scout bee behavior introduced in the refined version of Szeto et al. (2011) for this subsection's tests.

After removing the modifications from our ABC heuristic, we randomly selected five benchmark instances and ran the original ABC (OABC) algorithm five times for each instance. All of the parameter settings were kept the same, as shown in Table 1. However, parameters sb, ρ, and μ were removed from OABC, and the new parameter *limit* was set to $(n + m - 1) \times 10$.

In Table 3, the third column gives the average objective value in five runs, and the fourth column gives the computation time in seconds when the best objective value is found for OABC. The fifth and sixth columns give the average objective value in five runs, and the computation time is given in seconds when the best objective value is found for ABC.

According to the results given in Table 3, one can obviously see that our food source acceptance criterion and scout bee search behavior significantly contribute to the solution quality. As far as the efficiency is concerned, the results show that the OABC algorithm very slightly outperforms our ABC algorithm. However, both algorithms find the solution within reasonable computational times.

Table 3
Comparison of the experimental results for different food source acceptance criteria.

Instance	m	OABC		ABC	
		Average	T (s)	Average	T (s)
c101	1	320	6.6	320	0.4
c101	2	584	18.5	590	12.3
c101	3	794	20.9	810	14.7
c101	4	968	30.9	1020	13.6
c101	10	1588	18.7	1786	59.1
c108	1	368	1.5	370	0.9
c108	2	674	11.3	680	3.1
c108	3	884	29.2	920	14
c108	4	1064	23.3	1126	24
c108	10	1676	28.7	1792	44.2
pro05	1	533	168.2	586.8	91.4
pro05	2	927.2	128.2	1045	220.7
pro05	3	1235	107.4	1429.8	400.2
pro05	4	1512.4	130.8	1755.4	330.7
pro05	15	3056.4	87	3351	331.1
pro20	1	547	637.1	604.8	346.9
pro20	2	1033	347.9	1173.3	600.5
pro20	3	1376	181.7	1587.8	677.6
pro20	4	1673	324.7	1958.8	826.3
r201	1	780.2	19.9	787	20.3
r201	2	1156.8	20.2	1231.2	23
r201	3	1366.4	27.5	1434.8	39.7
r201	4	1436.2	22.2	1458	1

5.4. Computational results

To compare our ABC method with the I3CH, GVNS, SSA, GRASP-ELS, VNS, ILS, and ACS algorithms, we ran it five times for each one of the TOPTW benchmark instances. However, SSA and ILS were executed only once; GVNS, GRASP-ELS, and ACS were executed for five runs; and VNS was executed for ten runs. The computational results are given in Tables 4–13.

In Tables 4–7 and 9–12, the first column describes the benchmark instances, and the second column gives the best known solution (BKS). The third, fourth, fifth, sixth, seventh, eighth, ninth, tenth, 11th, 12th, 13th, 14th, 15th, 16th, 17th, 18th, 19th, 20th, 21th, 22th, 27th, 28th and 29th columns are taken from the original paper of Labadie et al. (2012), Lin and Yu (2012) and Hu and Lim (2014), and they give the average solution profit obtained by the GVNS over five runs P_{avg}, the percentage gap between the best-known solution and the obtained solution, the average CPU time in terms of seconds required by the GVNS heuristic for each run, the average solution profit obtained by the VNS heuristic over the ten runs P_{avg}, the percentage gap between the best-known solution and the obtained solution, the average CPU time in terms of seconds required by the VNS heuristic for each run, the best profit obtained by the SSA heuristic P, the gap between the best-known solution and the obtained solution, the computation time in seconds required by the SSA heuristic, the best profit obtained by the ILS heuristic, the percentage gap between the best-known solution and the obtained solution, the CPU time in terms of

Table 4
Results for the test problems of Solomon (m = 1).

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)
c101	320.0	320	0.0	0.2	320	0.0	72.2	320	0.0	20.4	320.0	0.0	1.2
c102	360.0	360	0.0	67.4	360	0.0	102.2	360	0.0	20.7	360.0	0.0	16.5
c103	400.0	396	1.0	382.4	398	0.5	106.2	400	0.0	24.2	400.0	0.0	38.5
c104	420.0	410	2.4	1005.9	418	0.5	124.3	420	0.0	21.9	420.0	0.0	105.2
c105	340.0	340	0.0	3.3	340	0.0	87.1	340	0.0	20.6	340.0	0.0	2.3
c106	340.0	340	0.0	6.5	340	0.0	86.7	340	0.0	20.3	340.0	0.0	3.8
c107	370.0	358	3.4	0.7	370	0.0	88.7	370	0.0	20.5	370.0	0.0	3.7
c108	370.0	354	4.5	1.1	370	0.0	99.3	370	0.0	20.5	370.0	0.0	6.6
c109	380.0	380	0.0	30.6	380	0.0	118.7	380	0.0	20.5	380.0	0.0	25.5
c201	870.0	850	2.4	0.1	870	0.0	507.8	870	0.0	28.3	870.0	0.0	2.5
c202	930.0	916	1.5	78.7	928	0.2	454.8	930	0.0	33.5	910.0	2.2	20.4
c203	960.0	956	0.4	329	960	0.0	614.3	960	0.0	59.6	960.0	0.0	45.2
c204	972.0	966	0.6	974	972	0.0	484	970	0.2	42.3	970.0	0.2	130.3
c205	910.0	898	1.3	12.5	908	0.2	645.4	910	0.0	46.9	906.0	0.4	10.6
c206	930.0	922	0.9	34.4	927	0.3	616.5	930	0.0	29	926.0	0.4	13.4
c207	930.0	928	0.2	36.3	930	0.0	599.5	930	0.0	29.1	930.0	0.0	17.3
c208	950.0	942	0.8	74.2	949	0.1	558.9	950	0.0	31.2	942.0	0.8	17.7
r101	198.0	197	0.5	0.2	198	0.0	49.9	198	0.0	19.7	198.0	0.0	0.9
r102	286.0	274.8	4.1	13.4	285.2	0.3	77.9	286	0.0	21	286.0	0.0	2.3
r103	293.0	286	2.4	33.8	293	0.0	90.2	293	0.0	20.3	290.4	0.9	4.8
r104	303.0	298.6	1.5	72.7	303	0.0	95.9	303	0.0	23.2	302.8	0.1	5.1
r105	247.0	230.6	7.1	0.3	247	0.0	85.4	247	0.0	20.3	247.0	0.0	1.2
r106	293.0	280.4	4.5	19.1	292.2	0.3	89.6	293	0.0	22.1	293.0	0.0	3.3
r107	299.0	287.2	4.1	50.3	299	0.0	105.5	297	0.7	21.5	296.4	0.9	4.8
r108	308.0	301.4	2.2	50.1	308	0.0	119.1	306	0.7	40.6	306.8	0.4	5.0
r109	277.0	276.4	0.2	10.4	277	0.0	77.4	277	0.0	20.4	277.0	0.0	2.7
r110	284.0	279.2	1.7	16.8	284	0.0	86	284	0.0	20.8	283.6	0.1	2.8
r111	297.0	290.6	2.2	54.2	297	0.0	95.4	297	0.0	27.9	297.0	0.0	4.6
r112	298.0	289.6	2.9	31.9	297.9	0.0	97	298	0.0	22.3	297.2	0.3	4.6
r201	796.7	775.6	2.7	6.7	796.7	0.0	1021.7	794	0.3	51.1	788.2	1.1	5.3
r202	930.0	881.4	5.5	13.4	905.6	2.7	1057.5	914	1.8	46.4	909.8	2.2	9.6
r203	1020.0	992.2	2.8	34.7	1007.9	1.2	1139.9	997	2.3	44.2	1001.4	1.9	12.6
r204	1076.3	1073.8	0.2	77.6	1076.3	0.0	1253.5	1058	1.7	39.7	1071.8	0.4	15.2
r205	953.0	905.8	5.2	15.3	952.6	0.0	745.1	946	0.7	37.8	927.2	2.8	7.8
r206	1023.6	966.6	5.9	34.9	1014	0.9	1168.1	1020	0.4	40.9	1023.6	0.0	10.7
r207	1069.0	1022	4.6	46.6	1061.5	0.7	1065.3	1069	0.0	52.5	1049.2	1.9	13.5
r208	1101.5	1083.6	1.7	52.9	1101.5	0.0	1160.8	1079	2.1	35.8	1100.2	0.1	15.9
r209	948.0	926	2.4	37.3	947.8	0.0	925.9	945	0.3	61.6	937.4	1.1	10.3
r210	982.0	961.4	2.1	30.1	975.9	0.6	1065.6	973	0.9	53.6	969.0	1.3	10.8
r211	1041.0	1025.6	1.5	22.5	1024.2	1.6	1120.7	1041	0.0	40.5	1028.2	1.2	11.3
rc101	219.0	219	0.0	2.1	219	0.0	61.3	219	0.0	19.8	219.0	0.0	0.6
rc102	266.0	246	8.1	5.8	266	0.0	53.3	266	0.0	20.2	259.0	2.7	2.1
rc103	266.0	253.2	5.1	23.3	266	0.0	62.9	266	0.0	20.7	265.2	0.3	2.7
rc104	301.0	301	0.0	16.2	301	0.0	58	301	0.0	27.5	300.2	0.3	3.1
rc105	244.0	227	7.5	2.7	244	0.0	78.6	244	0.0	20.3	244.0	0.0	1.4
rc106	252.0	252	0.0	2.2	252	0.0	71.9	252	0.0	21	252.0	0.0	1.2
rc107	277.0	261.2	6.0	15.7	277	0.0	69.6	277	0.0	22	277.0	0.0	1.9
rc108	298.0	288.8	3.2	10.4	297	0.3	66.1	298	0.0	26	298.0	0.0	2.9
rc201	795.0	784	1.4	3.7	795	0.0	640.8	795	0.0	44.4	786.4	1.1	4.2
rc202	930.0	890.6	4.4	12.1	925.6	0.5	951.5	930	0.0	46.3	922.8	0.8	8.9
rc203	988.2	954.4	3.5	22.5	988.2	0.0	938.4	967	2.2	32.4	978.2	1.0	9.6
rc204	1140.0	1101.8	3.5	31.9	1120.8	1.7	970.3	1140	0.0	46.5	1096.0	4.0	11.9
rc205	854.0	843.8	1.2	7	845.9	1.0	726.6	854	0.0	52.6	844.6	1.1	6.8
rc206	890.2	866.6	2.7	11.6	878.4	1.3	838.7	885	0.6	60.2	886.8	0.4	6.1
rc207	977.0	911.4	7.2	14.7	960.8	1.7	893.5	977	0.0	68.4	965.4	1.2	7.5
rc208	1043.5	1000.2	4.3	24.6	1043.5	0.0	995.5	1041	0.2	51.2	1007.0	3.6	10.7
Average		606.1	2.6	70.9	619.0	0.3	457.8	619.2	0.3	33.1	616.0	0.7	12.6

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	$P_{\max}$	P_{avg}	Gap (%)	T (s)
c101	320.0	320.0	0.0	0.4	320.0	0.0	0.5	320.0	0.0	0.4	320.0	320.0	0.0	21.8
c102	360.0	360.0	0.0	0.3	360.0	0.0	0.7	360.0	0.0	0.6	360.0	360.0	0.0	28.1
c103	400.0	390.0	2.6	0.5	400.0	0.0	16.9	398.0	0.5	3.6	399.0	400.0	0.0	27.1
c104	420.0	400.0	5.0	0.3	420.0	0.0	33.5	420.0	0.0	15.2	420.0	420.0	0.0	27.1
c105	340.0	340.0	0.0	0.3	340.0	0.0	0.9	340.0	0.0	0.5	340.0	340.0	0.0	23.4
c106	340.0	340.0	0.0	0.3	340.0	0.0	1.0	340.0	0.0	0.5	340.0	340.0	0.0	23.6
c107	370.0	360.0	2.8	0.3	370.0	0.0	2.1	370.0	0.0	1.3	370.0	370.0	0.0	24.7
c108	370.0	370.0	0.0	0.3	370.0	0.0	0.8	370.0	0.0	0.9	370.0	370.0	0.0	24.8
c109	380.0	380.0	0.0	0.3	380.0	0.0	0.8	380.0	0.0	0.5	380.0	380.0	0.0	26.3
c201	870.0	840.0	3.6	1.1	870.0	0.0	507.8	870.0	0.0	11.2	870.0	870.0	0.0	70.1
c202	930.0	910.0	2.2	2.8	928.0	0.2	454.8	930.0	0.0	15.3	930.0	930.0	0.0	87.6
c203	960.0	940.0	2.1	1.7	960.0	0.0	614.3	958.0	0.2	14.8	960.0	960.0	0.0	92.3

(continued on next page)

Table 4 (*continued*)

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{max}	P_{avg}	Gap (%)	T (s)
c204	972.0	950.0	2.3	1.6	972.0	0.0	484.0	966.0	0.6	25.7	971.0	970.0	0.2	117.4
c205	910.0	900.0	1.1	1.2	908.0	0.2	645.4	900.0	1.1	1.7	904.0	900.0	1.1	70.7
c206	930.0	910.0	2.2	1.6	927.0	0.3	616.5	928.0	0.2	6.2	930.0	920.0	1.1	75.7
c207	930.0	910.0	2.2	2.1	930.0	0.0	599.5	918.0	1.3	2.3	920.0	930.0	0.0	77.4
c208	950.0	930.0	2.2	1.6	949.0	0.1	558.9	950.0	0.0	11.8	950.0	950.0	0.0	84.0
r101	198.0	182.0	8.8	0.1	198.0	0.0	0.1	190.0	4.2	0.3	191.0	198	0.0	20.4
r102	286.0	286.0	0.0	0.2	286.0	0.0	11.1	281.6	1.6	2.2	282.0	286.0	0.0	29.3
r103	293.0	286.0	2.4	0.2	292.6	0.1	640.6	292.0	0.3	1.1	292.0	293.0	0.0	28.8
r104	303.0	297.0	2.0	0.2	303.0	0.0	164.0	299.6	1.1	1.7	301.0	298.0	1.7	27.3
r105	247.0	247.0	0.0	0.1	247.0	0.0	3.0	242.6	1.8	9.1	245.0	247.0	0.0	26.0
r106	293.0	293.0	0.0	0.2	293.0	0.0	86.3	289.0	1.4	1.6	290.0	293.0	0.0	29.4
r107	299.0	288.0	3.8	0.2	294.6	1.5	922.6	299.0	0.0	7.1	299.0	297.0	0.7	27.8
r108	308.0	297.0	3.7	0.2	306.0	0.7	696.1	308.0	0.0	4.6	308.0	306.0	0.7	29.7
r109	277.0	276.0	0.4	0.2	277.0	0.0	28.0	277.0	0.0	1.6	277.0	277.0	0.0	31.1
r110	284.0	281.0	1.1	0.3	283.2	0.3	617.6	282.2	0.6	8.1	284.0	284.0	0.0	33.9
r111	297.0	295.0	0.7	0.2	296.6	0.1	484.4	294.0	1.0	1.7	297.0	295.0	0.7	27.7
r112	298.0	295.0	1.0	0.2	297.4	0.2	947.1	297.0	0.3	2.4	298.0	289.0	3.1	32.0
r201	796.7	788.0	1.1	1.2	796.7	0.0	1021.7	787.0	1.2	20.3	788.0	789	1.0	101.8
r202	930.0	880.0	5.7	1.4	905.6	2.7	1057.5	895.8	3.8	40.6	904.0	930.0	0.0	175.6
r203	1020.0	980.0	4.1	1.6	1007.9	1.2	1139.9	1009.0	1.1	25.6	1016.0	1020.0	0.0	221.4
r204	1076.3	1073.0	0.3	1.7	1076.3	0.0	1253.5	1070.4	0.6	34.0	1073.0	1073.0	0.3	236.9
r205	953.0	931.0	2.4	1.4	952.6	0.0	745.1	953.0	0.0	32.9	953.0	946.0	0.7	129.3
r206	1023.6	996.0	2.8	1.5	1014.0	0.9	1168.1	1018.0	0.6	21.3	1021.0	1021.0	0.3	169.3
r207	1069.0	1038.0	3.0	2.0	1061.5	0.7	1065.3	1060.8	0.8	20.9	1069.0	1050.0	1.8	192.8
r208	1101.5	1069.0	3.0	1.6	1101.5	0.0	1160.8	1084.6	1.6	26.7	1094.0	1092.0	0.9	230.0
r209	948.0	926.0	2.4	2.4	947.8	0.0	925.9	934.6	1.4	26.3	944.0	948.0	0.0	136.5
r210	982.0	958.0	2.5	1.9	975.9	0.6	1065.6	965.4	1.7	29.3	966.0	982.0	0.0	176.9
r211	1041.0	1023.0	1.8	1.6	1024.2	1.6	1120.7	1019.0	2.2	15.0	1023.0	1013.0	2.8	167.4
rc101	219.0	219.0	0.0	0.2	219.0	0.0	0.2	219.0	0.0	0.5	219.0	219	0.0	21.8
rc102	266.0	259.0	2.7	0.2	266.0	0.0	30.9	266.0	0.0	0.6	266.0	266.0	0.0	25.5
rc103	266.0	265.0	0.4	0.3	266.0	0.0	57.2	266.0	0.0	10.7	266.0	266.0	0.0	27.1
rc104	301.0	297.0	1.3	0.3	301.0	0.0	29.4	301.0	0.0	20.3	301.0	301.0	0.0	27.2
rc105	244.0	221.0	10.4	0.2	244.0	0.0	9.7	244.0	0.0	1.3	244.0	244.0	0.0	26.4
rc106	252.0	239.0	5.4	0.2	252.0	0.0	308.6	251.4	0.2	8.1	252.0	250.0	0.8	25.0
rc107	277.0	274.0	1.1	0.2	277.0	0.0	502.1	276.2	0.3	10.4	277.0	274.0	1.1	26.3
rc108	298.0	288.0	3.5	0.2	298.0	0.0	207.5	298.0	0.0	11.3	298.0	264.0	12.9	25.1
rc201	795.0	780.0	1.9	1.0	795.0	0.0	640.8	784.0	1.4	24.9	789.0	795	0.0	80.9
rc202	930.0	882.0	5.4	1.3	925.6	0.5	951.5	926.8	0.3	27.3	931.0	924.0	0.6	129.3
rc203	988.2	960.0	2.9	2.7	988.2	0.0	938.4	962.4	2.7	24.8	967.0	966.0	2.3	134.3
rc204	1140.0	1117.0	2.1	2.3	1120.8	1.7	970.3	1109.2	2.8	23.4	1116.0	1093.0	4.3	167.5
rc205	854.0	840.0	1.7	1.0	845.9	1.0	726.6	852.3	0.2	27.2	854.0	847.0	0.8	99.2
rc206	890.2	860.0	3.5	1.1	878.4	1.3	838.7	890.2	0.0	24.9	895.0	863.0	3.2	98.4
rc207	977.0	926.0	5.5	1.3	960.8	1.7	893.5	977.0	0.0	15.8	977.0	957.0	2.1	122.0
rc208	1043.5	1037.0	0.6	2.3	1043.5	0.0	995.5	1032.8	1.0	26.3	1042.0	1003.0	4.0	123.0
Average		607.1	2.3	0.9	619.0	0.3	517.2	616.5	0.7	12.6	618.6	624.8	0.9	79.2

Table 5
Results for the test problems of Solomon ($m = 2$).

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)
c101	590.0	590	0.0	9.3	587	0.5	83.8	590	0.0	26.2	590.0	0.0	3.0
c102	660.0	648	1.9	47.5	653	1.1	100	660	0.0	26.5	656.0	0.6	34.1
c103	720.0	704	2.3	271.6	717	0.4	93.8	720	0.0	26.8	720.0	0.0	190.9
c104	760.0	746	1.9	616.4	759	0.1	96.7	760	0.0	28	760.0	0.0	319.7
c105	640.0	640	0.0	35	640	0.0	68.9	640	0.0	26.1	640.0	0.0	4.6
c106	620.0	620	0.0	44.4	620	0.0	95.8	620	0.0	25.6	620.0	0.0	10.2
c107	670.0	670	0.0	34.8	669	0.1	92.5	670	0.0	26.3	670.0	0.0	8.9
c108	680.0	680	0.0	65.7	680	0.0	89.7	680	0.0	26.2	680.0	0.0	17.7
c109	720.0	716	0.6	131.1	719	0.1	70.6	720	0.0	26.1	720.0	0.0	49.4
c201	1452.0	1450	0.1	5.8	1447	0.3	634.5	1450	0.1	64.3	1450.0	0.1	4.1
c202	1470.0	1464	0.4	21.1	1457	0.9	658.7	1460	0.7	50.2	1470.0	0.0	17.7
c203	1472.0	1464	0.5	46.1	1471	0.1	568.9	1450	1.5	39.4	1472.0	0.0	45.0
c204	1480.0	1472	0.5	135.5	1480	0.0	420.3	1450	2.1	38.9	1480.0	0.0	94.4
c205	1470.0	1466	0.3	10.8	1457	0.9	593.4	1470	0.0	68	1470.0	0.0	10.9
c206	1480.0	1472	0.5	14.1	1464	1.1	509.4	1460	1.4	42.4	1480.0	0.0	17.1
c207	1484.0	1484	0.0	18.1	1469	1.0	496.6	1480	0.3	87.7	1482.0	0.1	19.9
c208	1486.0	1480	0.4	18.8	1473	0.9	483.3	1460	1.8	38.4	1486.0	0.0	25.0
r101	349.0	349	0.0	4	349	0.0	35.3	349	0.0	25	346.0	0.9	1.6

Table 5 (*continued*)

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)
r102	508.0	492.8	3.1	17.4	498.5	1.9	64.1	508	0.0	43.9	505.2	0.6	6.5
r103	519.0	506.6	2.4	37.4	515.4	0.7	51	519	0.0	32.3	512.2	1.3	11.1
r104	549.0	539.2	1.8	93.7	542.5	1.2	66.9	548	0.2	29.1	543.8	1.0	12.1
r105	453.0	442.6	2.3	5.3	451.5	0.3	66.8	453	0.0	25.4	447.0	1.3	2.5
r106	529.0	524.4	0.9	31.2	518.1	2.1	66.4	529	0.0	27.2	512.4	3.2	7.5
r107	533.0	522.2	2.1	56.5	529	0.8	73.9	532	0.2	37	526.8	1.2	10.0
r108	558.0	545.6	2.3	179.5	547.3	2.0	63.7	558	0.0	67.5	553.4	0.8	12.8
r109	506.0	501.6	0.9	19.7	505	0.2	66.3	506	0.0	45.5	504.8	0.2	5.3
r110	525.0	523	0.4	65.5	509.4	3.1	62.5	525	0.0	27.5	507.4	3.5	6.4
r111	544.0	530.6	2.5	115.1	537.4	1.2	71.4	544	0.0	54	531.0	2.4	10.5
r112	544.0	536.6	1.4	98.8	535.2	1.6	73.2	542	0.4	25.1	528.4	3.0	9.3
r201	1242.0	1225.8	1.3	15	1232.1	0.8	1401.6	1242	0.0	73.7	1236.8	0.4	14.4
r202	1344.0	1306.2	2.9	15.4	1333.5	0.8	1421.5	1334	0.7	50.4	1339.2	0.4	21.4
r203	1416.0	1392.6	1.7	17.7	1402.7	0.9	1148.4	1414	0.1	196.2	1409.6	0.5	23.4
r204	1458.0	1452.6	0.4	8.9	1455.3	0.2	785.2	1447	0.8	70.9	1456.2	0.1	9.6
r205	1380.0	1351.6	2.1	20.1	1366.8	1.0	1276	1363	1.2	51.5	1365.0	1.1	18.0
r206	1430.0	1429.8	0.0	17.1	1422.4	0.5	1015.9	1430	0.0	113.7	1428.0	0.1	24.0
r207	1458.0	1449.8	0.6	12.4	1453.6	0.3	723.2	1452	0.4	158.8	1457.2	0.1	16.4
r208	1458.0	1458	0.0	8.3	1458	0.0	448.7	1457	0.1	56	1458.0	0.0	0.1
r209	1404.0	1389.6	1.0	16.3	1393	0.8	1076	1404	0.0	89.5	1384.4	1.4	23.4
r210	1415.0	1391	1.7	16	1408.6	0.5	912	1414	0.1	77.8	1408.8	0.4	20.4
r211	1457.0	1452.6	0.3	14.8	1457	0.0	957.5	1451	0.4	66.9	1455.2	0.1	22.3
rc101	427.0	420.6	1.5	2.8	419	1.9	54.9	427	0.0	25.3	414.0	3.1	1.5
rc102	505.0	485.4	4.0	8.4	504.1	0.2	61.1	505	0.0	42.4	484.6	4.2	4.4
rc103	523.0	502	4.2	26.8	519	0.8	53.1	523	0.0	26.5	510.4	2.5	6.8
rc104	575.0	554.8	3.6	43.6	568.2	1.2	53	575	0.0	63.1	555.8	3.5	7.1
rc105	480.0	456.8	5.1	7.1	479.8	0.0	45.3	480	0.0	56.3	475.8	0.9	2.9
rc106	483.0	480.4	0.5	11	477.9	1.1	57.5	483	0.0	47.3	472.0	2.3	3.7
rc107	531.4	524	1.4	26	519.3	2.3	63.2	529	0.5	26.8	527.0	0.8	4.6
rc108	554.0	544.6	1.7	36.8	535.8	3.4	53.2	554	0.0	36.1	550.2	0.7	6.3
rc201	1377.0	1359.6	1.3	7.3	1359.7	1.3	897.1	1377	0.0	73.3	1370.0	0.5	11.1
rc202	1502.4	1465	2.6	16.8	1487.6	1.0	909.1	1499	0.2	51.8	1502.4	0.0	14.9
rc203	1627.0	1561.6	4.2	11.8	1593.4	2.1	821	1576	3.2	94.9	1606.4	1.3	19.5
rc204	1710.2	1699.2	0.6	8.8	1710.2	0.0	740.8	1674	2.2	115.6	1701.2	0.5	22.1
rc205	1458.0	1396	4.4	8.5	1436.3	1.5	766.8	1458	0.0	50.5	1446.6	0.8	14.6
rc206	1528.0	1507.4	1.4	18.1	1523.4	0.3	769.6	1528	0.0	50.8	1511.4	1.1	16.0
rc207	1582.0	1546.6	2.3	18.3	1552.2	1.9	835.9	1574	0.5	73.5	1569.6	0.8	18.2
rc208	1676.1	1669.6	0.4	12.5	1676.1	0.0	698.2	1675	0.1	130.4	1667.4	0.5	20.7
Average		986.7	1.4	47.8	991.9	0.8	427.9	994.6	0.3	54.9	993.4	0.9	23.9

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	$P_{\max}$	P_{avg}	Gap (%)	T (s)
c101	590.0	590	0.0	1.4	588	0.3	110.8	590	0.0	12.3	590	590	0.0	53.4
c102	660.0	650	1.5	0.9	660	0.0	1427.4	660	0.0	12.4	660	660	0.0	78.4
c103	720.0	700	2.9	1.2	710	1.4	938.9	716	0.6	19.6	720	720	0.0	116.1
c104	760.0	750	1.3	1.5	754	0.8	1355	758	0.3	13.5	760	760	0.0	94.4
c105	640.0	640	0.0	0.8	640	0.0	1436.6	640	0.0	2.1	640	640	0.0	70.6
c106	620.0	620	0.0	0.8	620	0.0	104.6	620	0.0	2.3	620	620	0.0	149.2
c107	670.0	670	0.0	1.4	670	0.0	76	670	0.0	2.1	670	670	0.0	65.4
c108	680.0	670	1.5	0.8	680	0.0	95.3	680	0.0	3.1	680	680	0.0	85.9
c109	720.0	710	1.4	0.9	720	0.0	1817.3	720	0.0	8.9	720	720	0.0	69.4
c201	1452.0	1400	3.7	2.7	1452	0.0	508.6	1450	0.1	22.4	1460	1450	0.1	321.2
c202	1470.0	1430	2.8	5.1	1446	1.7	1568.2	1458	0.8	32.2	1460	1470	0.0	405.9
c203	1472.0	1430	2.9	3.8	1446	1.8	2334.8	1450	1.5	15.9	1457	1470	0.1	458.2
c204	1480.0	1460	1.4	4.2	1436	3.1	1373.2	1450	2.1	17.5	1460	1480	0.0	498.5
c205	1470.0	1450	1.4	3.1	1452	1.2	925.4	1470	0.0	24.7	1470	1450	1.4	322.2
c206	1480.0	1440	2.8	2.8	1460	1.4	1876.4	1470	0.7	3.8	1470	1480	0.0	355.9
c207	1484.0	1450	2.3	3.2	1452	2.2	1434	1477	0.5	28	1480	1470	1.0	423.5
c208	1486.0	1460	1.8	2.8	1462	1.6	1164.2	1472	1.0	9.2	1479	1470	1.1	424.3
r101	349.0	330	5.8	0.4	349	0.0	36.5	342.8	1.8	2.9	345	349	0.0	42.1
r102	508.0	508	0.0	0.9	506.6	0.3	2006.6	508	0.0	26.1	508	508	0.0	62.4
r103	519.0	513	1.2	0.9	518.8	0.0	1095.1	517.2	0.3	19.1	519	519	0.0	68.3
r104	549.0	539	1.9	1.5	540	1.7	2291.8	549	0.0	17.3	549	549	0.0	75.3
r105	453.0	430	5.3	0.8	453	0.0	1037.3	448	1.1	20.7	450	447	1.3	56.2
r106	529.0	529	0.0	0.9	523.8	1.0	2034.7	529	0.0	11.1	529	529	0.0	63.5
r107	533.0	529	0.8	1	527.4	1.1	2357.6	518.8	2.7	21.9	520	533	0.0	63.2
r108	558.0	549	1.6	1.4	553.6	0.8	1906.2	556	0.4	13.7	557	550	1.5	65.7
r109	506.0	498	1.6	0.5	505.4	0.1	1386.6	502.8	0.6	8.9	504	506	0.0	60.5
r110	525.0	515	1.9	1	523.4	0.3	1180.2	525	0.0	12.4	525	525	0.0	68.9
r111	544.0	535	1.7	0.6	535.6	1.6	1726.2	544	0.0	11.8	544	542	0.4	67
r112	544.0	515	5.6	0.5	541.6	0.4	1653.6	544	0.0	11.5	544	534	1.9	63
r201	1242.0	1231	0.9	2.1	1236	0.5	2693.6	1231.2	0.9	23	1239	1254	−1.0	333.6

(*continued on next page*)

Table 5 (*continued*)

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	$P_{\max}$	P_{avg}	Gap (%)	T (s)
r202	1344.0	1270	5.8	2.3	1298.6	3.5	2482.2	1341.6	0.2	40.2	1344	1344	0.0	588.5
r203	1416.0	1377	2.8	1.9	1349.4	4.9	3057.4	1410	0.4	43.7	1414	1416	0.0	815.9
r204	1458.0	1440	1.3	3.4	1399	4.2	2713.5	1453.4	0.3	35.6	1457	1458	0.0	33.5
r205	1380.0	1338	3.1	2.8	1342	2.8	2593.7	1365.8	1.0	47.1	1375	1380	0.0	606.1
r206	1430.0	1401	2.1	2.8	1373.8	4.1	2954.4	1426.8	0.2	48.4	1430	1427	0.2	739.9
r207	1458.0	1428	2.1	1.7	1385	5.3	2624.1	1448.4	0.7	35.7	1455	1458	0.0	453.8
r208	1458.0	1458	0.0	1.6	1425.4	2.3	2828.9	1457.8	0.0	12.6	1458	1458	0.0	4.3
r209	1404.0	1345	4.4	2.6	1350.6	4.0	2988.9	1390.2	1.0	35	1400	1404	0.0	600.3
r210	1415.0	1365	3.7	1.9	1355.4	4.4	2423.9	1414.2	0.1	38.2	1419	1415	0.0	728.5
r211	1457.0	1422	2.5	1.9	1403	3.8	2726.1	1451.2	0.4	29.3	1457	1450	0.5	890.9
rc101	427.0	427	0.0	0.6	427	0.0	27.9	425.8	0.3	5.4	427	427	0.0	52.3
rc102	505.0	494	2.2	0.8	497.2	1.6	1496.9	505	0.0	5.1	505	505	0.0	59.8
rc103	523.0	519	0.8	1.1	510.2	2.5	1965.9	523	0.0	8.7	524	519	0.8	60.3
rc104	575.0	565	1.8	0.7	568.8	1.1	2381.3	572.8	0.4	26.2	575	556	3.4	59.9
rc105	480.0	459	4.6	0.8	478.4	0.3	1676.7	475.2	1.0	28.2	476	480	0.0	58.6
rc106	483.0	458	5.5	0.6	480.2	0.6	1865.1	483	0.0	12.6	483	481	0.4	56
rc107	531.4	515	3.2	0.5	530.2	0.2	771.5	531.4	0.0	19.8	534	529	0.5	62.9
rc108	554.0	546	1.5	0.6	541.6	2.3	821	554	0.0	17.3	556	547	1.3	61.4
rc201	1377.0	1305	5.5	1.9	1365.4	0.8	1654.3	1373.2	0.3	18.5	1377	1384	−0.5	267.4
rc202	1502.4	1461	2.8	2.1	1464.8	2.6	2619.6	1472.6	2.0	36.8	1475	1500	0.2	386.7
rc203	1627.0	1573	3.4	2	1542.2	5.5	2176.2	1593.6	2.1	42	1594	1627	0.0	482.8
rc204	1710.2	1656	3.3	2.1	1614.8	5.9	2561.8	1682	1.7	49.6	1688	1704	0.4	760.8
rc205	1458.0	1381	5.6	3.2	1417.4	2.9	2115.2	1446	0.8	26.2	1449	1452	0.4	311
rc206	1528.0	1495	2.2	1.9	1495.6	2.2	2105.1	1503.8	1.6	28.1	1506	1525	0.2	335.9
rc207	1582.0	1531	3.3	2.7	1523	3.9	2937.5	1561.8	1.3	37.7	1567	1582	0.0	408.6
rc208	1676.1	1606	4.4	1.7	1608.4	4.2	2572.3	1647.4	1.7	46.1	1653	1669	0.4	564.4
Average		974.6	2.4	1.7	977.0	1.8	1733.8	992.4	0.6	21.5	995.1	1008.2	0.3	266.1

Table 6
Results for the test problems of Solomon ($m = 3$).

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)
c101	810.0	808	0.2	18	810	0.0	67	810	0.0	31.7	810	0.0	4.6
c102	920.0	916	0.4	94.1	916	0.4	89.3	920	0.0	32.7	920	0.0	33.1
c103	990.0	964	2.7	332.3	972	1.9	91.4	980	1.0	37.3	980	1.0	161.3
c104	1030.0	1012	1.8	604	1020	1.0	78.9	1010	2.0	33.1	1028	0.2	462.9
c105	870.0	864	0.7	36.5	858	1.4	93.8	870	0.0	32.9	854	1.9	6.6
c106	870.0	866	0.5	50.2	867	0.3	89.5	870	0.0	39.6	864	0.7	13.5
c107	910.0	906	0.4	63.4	905	0.6	91.8	910	0.0	33.4	910	0.0	13.2
c108	920.0	914	0.7	105.9	913	0.8	85.3	920	0.0	33.1	910	1.1	24.2
c109	970.0	958	1.3	180.7	967	0.3	82.3	970	0.0	43.5	968	0.2	61.3
c201	1810.0	1800	0.6	3.7	1810	0.0	256.5	1800	0.6	48.1	1810	0.0	0
c202	1810.0	1806	0.2	8.8	1810	0.0	321.5	1790	1.1	68.3	1810	0.0	0.1
c203	1810.0	1804	0.3	10.9	1810	0.0	185.4	1770	2.3	111.9	1810	0.0	0.2
c204	1810.0	1802	0.4	10.9	1810	0.0	201.6	1750	3.4	54.5	1810	0.0	0.2
c205	1810.0	1810	0.0	4.9	1809	0.1	236.6	1810	0.0	42.5	1810	0.0	0
c206	1810.0	1810	0.0	6.5	1810	0.0	265.9	1780	1.7	40.8	1810	0.0	0
c207	1810.0	1810	0.0	7.6	1810	0.0	233.3	1800	0.6	59.5	1810	0.0	0
c208	1810.0	1810	0.0	8.3	1810	0.0	231.8	1800	0.6	52.2	1810	0.0	0
r101	484.0	477.4	1.4	7.7	482.3	0.4	42.9	484	0.0	48.6	478	1.3	2.7
r102	694.0	674.2	2.9	21.9	681.7	1.8	66.9	694	0.0	43.5	680.8	1.9	9.9
r103	746.0	724.4	3.0	68.4	726.6	2.7	61.2	736	1.4	46.1	728	2.5	14.5
r104	777.0	762	2.0	103.7	759.7	2.3	60.2	777	0.0	71.2	768.8	1.1	25.3
r105	619.3	612.4	1.1	11	619.2	0.0	52.7	619	0.0	36.5	613.6	0.9	5.8
r106	729.0	713.4	2.2	47	717.4	1.6	65.7	729	0.0	38.6	713.6	2.2	12.9
r107	760.0	755	0.7	110.1	755.8	0.6	60.7	759	0.1	90.8	751	1.2	15.9
r108	797.0	770.4	3.5	187.9	781.2	2.0	70.9	789	1.0	81.8	784	1.7	24.2
r109	710.0	699.6	1.5	24.1	703.9	0.9	64	702	1.1	40.2	700.4	1.4	9.1
r110	736.0	716.2	2.8	69.8	721.9	2.0	63.5	734	0.3	36.8	727	1.2	12.2
r111	773.0	749.4	3.1	62.6	763.4	1.3	65	771	0.3	46.9	762.6	1.4	15.9
r112	776.0	751	3.3	173	760	2.1	69.2	776	0.0	91.8	760.6	2.0	17.9
r201	1438.4	1416.4	1.6	6.4	1428.2	0.7	814.4	1429	0.7	50.3	1438.4	0.0	19
r202	1458.0	1450.2	0.5	21	1453.7	0.3	519.7	1458	0.0	59	1458	0.0	7.9
r203	1458.0	1458	0.0	9	1458	0.0	257.3	1458	0.0	39.9	1458	0.0	0.2
r204	1458.0	1458	0.0	11.9	1458	0.0	211.7	1458	0.0	38.4	1458	0.0	0.1
r205	1458.0	1458	0.0	9	1458	0.0	296.3	1458	0.0	39.3	1458	0.0	0
r206	1458.0	1458	0.0	9	1458	0.0	240.3	1458	0.0	38.9	1458	0.0	0.1
r207	1458.0	1458	0.0	0.2	1458	0.0	220.1	1458	0.0	38.5	1458	0.0	0

Table 6 (*continued*)

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)
r208	1458.0	1458	0.0	0.1	1458	0.0	216.6	1458	0.0	38.4	1458	0.0	0
r209	1458.0	1458	0.0	0.2	1458	0.0	272	1458	0.0	40.1	1458	0.0	0
r210	1458.0	1458	0.0	9.7	1458	0.0	289.9	1458	0.0	40	1458	0.0	0.1
r211	1458.0	1458	0.0	0.2	1458	0.0	200.1	1458	0.0	38.6	1458	0.0	0
rc101	621.0	602.4	3.1	5.7	619.8	0.2	58.7	621	0.0	34.2	605.8	2.5	3.4
rc102	714.0	692	3.2	14	710.3	0.5	60.4	710	0.6	41.3	695.2	2.7	7.8
rc103	764.0	741.2	3.1	47	746.5	2.3	67.3	764	0.0	44.6	748.6	2.1	12.1
rc104	834.0	826	1.0	85.6	824.5	1.2	59.8	814	2.5	33	799.8	4.3	12.9
rc105	682.0	670.2	1.8	20.8	677.2	0.7	63.8	682	0.0	32	659.6	3.4	5.4
rc106	706.0	692.6	1.9	17.4	686.2	2.9	49.7	706	0.0	61.5	686.2	2.9	6.9
rc107	773.0	756.4	2.2	30.7	761.6	1.5	58.2	773	0.0	43.8	753.4	2.6	9.2
rc108	789.0	773	2.1	48.2	778	1.4	66.9	778	1.4	52	773.4	2.0	11.5
rc201	1689.4	1676	0.8	11.6	1689.4	0.0	786.6	1681	0.5	98.3	1686.8	0.2	19
rc202	1724.0	1702.8	1.2	10.5	1711.6	0.7	491.2	1714	0.6	48.3	1715.4	0.5	22.6
rc203	1724.0	1724	0.0	7.9	1724	0.0	313.9	1724	0.0	39.1	1724	0.0	1
rc204	1724.0	1724	0.0	0.1	1724	0.0	222.8	1724	0.0	38.3	1724	0.0	0.1
rc205	1719.0	1702.2	1.0	11.6	1696.7	1.3	573.2	1709	0.6	122.3	1703.2	0.9	22.5
rc206	1724.0	1724	0.0	8.7	1724	0.0	339.9	1724	0.0	46	1724	0.0	0.7
rc207	1724.0	1724	0.0	8.7	1724	0.0	295.1	1724	0.0	40.4	1724	0.0	0.8
rc208	1724.0	1724	0.0	0.2	1724	0.0	209.5	1724	0.0	39.1	1724	0.0	0
Average		1187.7	1.1	50.7	1191.2	0.7	191.1	1191.3	0.4	49.0	1190.3	0.9	19.8

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	$P_{\max}$	P_{avg}	Gap (%)	T (s)
c101	810.0	790	2.5	1.1	810	0.0	1515.6	810	0.0	14.7	810	810	0.0	303.1
c102	920.0	890	3.4	2.1	918	0.2	442.8	920	0.0	5.8	920	920	0.0	237.5
c103	990.0	960	3.1	2.2	972	1.9	1415.3	976	1.4	16.4	980	990	0.0	156.5
c104	1030.0	1010	2.0	1.3	1004	2.6	681.7	1008	2.2	31.5	1014	1030	0.0	155.9
c105	870.0	840	3.6	1	870	0.0	1010	868	0.2	14.3	870	870	0.0	110.2
c106	870.0	840	3.6	1.1	870	0.0	1132.3	870	0.0	16.5	870	870	0.0	227.4
c107	910.0	900	1.1	1.5	910	0.0	892.7	910	0.0	15.4	910	910	0.0	151.8
c108	920.0	900	2.2	1.2	912	0.9	971.2	920	0.0	14	920	920	0.0	212.6
c109	970.0	950	2.1	2	954	1.7	1327.5	962	0.8	18.2	966	960	1.0	157.2
c201	1810.0	1750	3.4	2.2	1810	0.0	259.2	1808	0.1	35.4	1810	1810	0.0	3.8
c202	1810.0	1750	3.4	2	1784	1.5	1603.2	1792	1.0	39.8	1793	1810	0.0	7.2
c203	1810.0	1760	2.8	2	1738	4.1	1039.9	1766	2.5	45.3	1774	1810	0.0	7.9
c204	1810.0	1780	1.7	1.5	1758	3.0	1025.8	1758	3.0	35.8	1762	1810	0.0	11.2
c205	1810.0	1770	2.3	2.5	1796	0.8	2007.3	1804	0.3	43.5	1810	1810	0.0	12.6
c206	1810.0	1770	2.3	1.5	1788	1.2	1862	1810	0.0	1.3	1810	1810	0.0	18
c207	1810.0	1810	0.0	3.4	1784	1.5	1223	1810	0.0	5.1	1810	1810	0.0	16.9
c208	1810.0	1810	0.0	2.4	1786	1.3	1542.7	1810	0.0	2	1810	1810	0.0	20.6
r101	484.0	481	0.6	0.8	481	0.6	240.6	477.2	1.4	11.4	480	481	0.6	67.7
r102	694.0	685	1.3	1	682	1.8	2435.4	690.6	0.5	19.4	691	691	0.4	111.4
r103	746.0	720	3.6	2	729.8	2.2	2694.9	746	0.0	39.2	747	740	0.8	125.6
r104	777.0	765	1.6	1.5	760.6	2.2	2531.4	777	0.0	25.7	777	777	0.0	128.3
r105	619.3	609	1.7	2.3	619.2	0.0	796	619.3	0.0	24.9	620	619	0.0	165.8
r106	729.0	719	1.4	2.1	715	2.0	755.5	725.8	0.4	24.5	726	729	0.0	122.7
r107	760.0	747	1.7	1.1	751.6	1.1	1998.9	760	0.0	24.2	760	759	0.1	120.5
r108	797.0	790	0.9	3.1	781.8	1.9	2388.5	785.8	1.4	27	792	797	0.0	135.7
r109	710.0	699	1.6	1.8	701.8	1.2	1098.8	704	0.9	25.9	710	710	0.0	103.6
r110	736.0	711	3.5	1.4	732.6	0.5	1708.5	734	0.3	29.7	737	736	0.0	109.9
r111	773.0	764	1.2	1.8	755.4	2.3	1286.7	771	0.3	20.8	773	773	0.0	118.1
r112	776.0	758	2.4	1.1	758.8	2.3	2091.2	770.8	0.7	28.2	772	776	0.0	110.5
r201	1438.4	1408	2.2	2.4	1428.4	0.7	1400	1434.8	0.3	39.7	1441	1439	0.0	981.2
r202	1458.0	1443	1.0	2.7	1445.6	0.9	1966.2	1458	0.0	5.8	1458	1458	0.0	15.3
r203	1458.0	1458	0.0	1.6	1452.4	0.4	2564.3	1458	0.0	0.8	1458	1458	0.0	0.2
r204	1458.0	1458	0.0	1	1458	0.0	174.7	1458	0.0	0.4	1458	1458	0.0	0.2
r205	1458.0	1458	0.0	1.1	1454.8	0.2	1830.7	1458	0.0	1.3	1458	1458	0.0	0.3
r206	1458.0	1458	0.0	1.1	1456.8	0.1	1328.8	1458	0.0	0.6	1458	1458	0.0	0.3
r207	1458.0	1458	0.0	1	1458	0.0	598.6	1458	0.0	0.4	1458	1458	0.0	0.3
r208	1458.0	1458	0.0	0.8	1458	0.0	425.6	1458	0.0	0.2	1458	1458	0.0	0.2
r209	1458.0	1458	0.0	1.1	1456.8	0.1	1163.7	1458	0.0	0.9	1458	1458	0.0	0.3
r210	1458.0	1458	0.0	1.2	1456.2	0.1	1428.1	1458	0.0	0.6	1458	1458	0.0	0.2
r211	1458.0	1458	0.0	1	1457.8	0.0	7.3	1458	0.0	0.4	1458	1458	0.0	0.4
rc101	621.0	604	2.8	1.4	620.4	0.1	195.7	617.4	0.6	23.5	621	621	0.0	87.6
rc102	714.0	698	2.3	1.3	698.8	2.2	1145.5	710.3	0.5	26.3	713	714	0.0	105.4
rc103	764.0	747	2.3	1.1	741.4	3.0	2754	748.6	2.1	16.5	754	764	0.0	105.7
rc104	834.0	822	1.5	1.3	813.2	2.6	2277	828	0.7	25.9	831	834	0.0	124.4
rc105	682.0	654	4.3	0.8	677.8	0.6	1092	682	0.0	28.7	682	682	0.0	90.4
rc106	706.0	678	4.1	1	692.4	2.0	1748.3	696.4	1.4	24.4	696	706	0.0	88.1
rc107	773.0	745	3.8	0.9	748	3.3	1158.5	769.8	0.4	34.7	772	762	1.4	98.7
rc108	789.0	757	4.2	1.1	768.2	2.7	1443.7	781.2	1.0	26.2	782	789	0.0	107.4

(*continued on next page*)

Table 6 (*continued*)

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	$P_{\max}$	P_{avg}	Gap (%)	T (s)
rc201	1689.4	1625	4.0	1.9	1672	1.0	1084.9	1681	0.5	32.1	1689	1693	−0.2	575.2
rc202	1724.0	1686	2.3	1.7	1701.4	1.3	2230.7	1717.4	0.4	12.7	1724	1724	0.0	1.1
rc203	1724.0	1724	0.0	2.9	1719.6	0.3	1213.2	1724	0.0	2.1	1724	1724	0.0	0.3
rc204	1724.0	1724	0.0	1	1722.2	0.1	1261.3	1724	0.0	1.1	1724	1724	0.0	0.3
rc205	1719.0	1659	3.6	2.4	1672.2	2.8	1439.8	1704.8	0.8	22.7	1709	1719	0.0	734.4
rc206	1724.0	1708	0.9	1.3	1712.6	0.7	2185.2	1724	0.0	4.3	1724	1724	0.0	0.5
rc207	1724.0	1713	0.6	1.5	1714.8	0.5	2553.9	1724	0.0	1.9	1724	1724	0.0	0.3
rc208	1724.0	1724	0.0	1.1	1722.2	0.1	893.9	1724	0.0	1.1	1724	1724	0.0	0.4
Average		1178.0	1.8	1.6	1184.2	1.2	1384.7	1191.7	0.5	17.7	1193.5	1206.9	0.1	89.2

Table 7
Results for the test problems of Solomon (m = 4).

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)
c101	1020.0	1014	0.6	22.1	1013	0.7	78.5	1020	0.0	37.1	1020	0.0	5.5
c102	1150.0	1132	1.6	58.2	1139	1.0	90.1	1150	0.0	40	1146	0.3	36.2
c103	1210.0	1178	2.7	221.2	1180	2.5	93.8	1190	1.7	38.1	1194	1.3	132.3
c104	1260.0	1226	2.8	513	1248	1.0	79.7	1230	2.4	41	1254	0.5	450.8
c105	1060.0	1060	0.0	33.3	1055	0.5	85	1060	0.0	36.8	1050	1.0	8.5
c106	1080.0	1052	2.7	51.5	1062	1.7	82.2	1080	0.0	69	1062	1.7	16.2
c107	1120.0	1112	0.7	62	1108	1.1	83.3	1120	0.0	45.5	1106	1.3	14.9
c108	1130.0	1116	1.3	94.5	1123	0.6	70.4	1130	0.0	69.1	1114	1.4	31.3
c109	1190.0	1170	1.7	143.2	1174	1.4	73.8	1190	0.0	69	1180	0.8	65.5
c201	1810.0	1810	0.0	0.1	1810	0.0	92.5	1810	0.0	44.9	1810	0.0	0
c202	1810.0	1810	0.0	1.9	1810	0.0	114.8	1810	0.0	44.2	1810	0.0	0
c203	1810.0	1810	0.0	1.5	1810	0.0	84.7	1810	0.0	42	1810	0.0	0.1
c204	1810.0	1810	0.0	0.5	1810	0.0	107	1810	0.0	39.6	1810	0.0	0
c205	1810.0	1810	0.0	0.1	1810	0.0	106.6	1810	0.0	42	1810	0.0	0
c206	1810.0	1810	0.0	0.1	1810	0.0	109.3	1810	0.0	40.6	1810	0.0	0
c207	1810.0	1810	0.0	0.1	1810	0.0	115	1810	0.0	40.8	1810	0.0	0
c208	1810.0	1810	0.0	0.1	1810	0.0	108.4	1810	0.0	40	1810	0.0	0
r101	611.0	607	0.7	11.9	610.2	0.1	40.4	611	0.0	33.8	600.2	1.8	4.4
r102	843.0	814.6	3.5	23	828.4	1.8	59.4	843	0.0	45.2	829.2	1.7	14.3
r103	928.0	902	2.9	68.9	909.9	2.0	68.8	926	0.2	97.1	917.6	1.1	24.6
r104	969.0	939	3.2	103	954.8	1.5	62.9	964	0.5	84.7	957.6	1.2	50.8
r105	778.0	770.6	1.0	15.5	768	1.3	62.5	771	0.9	47.3	766.4	1.5	9
r106	906.0	878.2	3.2	54.7	890.9	1.7	63.7	905	0.1	78.7	888.2	2.0	19.3
r107	950.0	932.8	1.8	112.4	930.4	2.1	62.6	942	0.8	42	938.6	1.2	29.6
r108	994.0	958.6	3.7	198.5	973.2	2.1	60.4	977	1.7	70.6	980	1.4	42.3
r109	885.0	869.4	1.8	55	870.5	1.7	55.3	885	0.0	68	865.8	2.2	13.4
r110	915.0	893.2	2.4	92.9	898.1	1.9	70.7	893	2.5	39.5	889.2	2.9	20.9
r111	952.0	937.4	1.6	96.5	936.6	1.6	63.8	948	0.4	53.2	942.8	1.0	24.6
r112	967.0	950.4	1.7	184.6	964.4	0.3	63.6	958	0.9	40.5	953.4	1.4	36.9
r201	1458.0	1458	0.0	1.4	1458	0.0	294.6	1458	0.0	42	1458	0.0	0.2
r202	1458.0	1458	0.0	0.1	1458	0.0	171.5	1458	0.0	41.6	1458	0.0	0
r203	1458.0	1458	0.0	0.2	1458	0.0	137.3	1458	0.0	40.2	1458	0.0	0.1
r204	1458.0	1458	0.0	0.1	1458	0.0	135.9	1458	0.0	38.9	1458	0.0	0
r205	1458.0	1458	0.0	0.1	1458	0.0	153.6	1458	0.0	39.6	1458	0.0	0
r206	1458.0	1458	0.0	0.2	1458	0.0	131.9	1458	0.0	39.4	1458	0.0	0
r207	1458.0	1458	0.0	0.2	1458	0.0	111.9	1458	0.0	38.7	1458	0.0	0
r208	1458.0	1458	0.0	0.1	1458	0.0	127	1458	0.0	38.1	1458	0.0	0
r209	1458.0	1458	0.0	0.2	1458	0.0	139	1458	0.0	39.6	1458	0.0	0
r210	1458.0	1458	0.0	0.2	1458	0.0	140.9	1458	0.0	40	1458	0.0	0
r211	1458.0	1458	0.0	0.2	1458	0.0	114.6	1458	0.0	38.7	1458	0.0	0
rc101	808.0	797.6	1.3	10.1	777.2	4.0	55.3	808	0.0	44.3	798	1.3	5.2
rc102	902.0	887	1.7	21.3	893.4	1.0	67	902	0.0	126.9	885	1.9	10.2
rc103	974.0	948.4	2.7	46.7	945.4	3.0	64.1	970	0.4	70	950	2.5	16.3
rc104	1064.0	1036.4	2.7	76.6	1033.5	3.0	64.8	1059	0.5	75.3	1033	3.0	21.9
rc105	875.0	866	1.0	20.3	859	1.9	53.6	875	0.0	46.2	854.4	2.4	10.3
rc106	909.0	896.2	1.4	20.7	894.4	1.6	49.6	901	0.9	41.4	887	2.5	9.2
rc107	980.0	963	1.8	46	958	2.3	53.8	980	0.0	93.8	962.6	1.8	14.1
rc108	1023.0	1014.8	0.8	53.6	1011.5	1.1	59.6	1023	0.0	47.1	1002.6	2.0	19.6
rc201	1724.0	1723	0.1	1.9	1724	0.0	231.9	1724	0.0	41.7	1724	0.0	0
rc202	1724.0	1724	0.0	0.8	1724	0.0	208.8	1724	0.0	41	1724	0.0	0
rc203	1724.0	1724	0.0	0.1	1724	0.0	157.5	1724	0.0	39.8	1724	0.0	0
rc204	1724.0	1724	0.0	0.1	1724	0.0	120.3	1724	0.0	38.7	1724	0.0	0

Table 7 (*continued*)

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)
rc205	1724.0	1724	0.0	3.3	1724	0.0	192.9	1724	0.0	41.6	1724	0.0	0.1
rc206	1724.0	1724	0.0	0.1	1724	0.0	145.5	1724	0.0	39.8	1724	0.0	0
rc207	1724.0	1724	0.0	0.5	1724	0.0	135.1	1724	0.0	39.7	1724	0.0	0.1
rc208	1724.0	1724	0.0	0.2	1724	0.0	124.3	1724	0.0	38.9	1724	0.0	0
Average		1289.9	1.0	45.1	1291.4	0.8	102.7	1296.8	0.3	49.7	1291.7	0.8	20.7

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	$P_{\max}$	P_{avg}	Gap (%)	T (s)
c101	1020.0	1000	2.0	3.8	1018	0.2	1049.1	1020	0.0	13.6	1020	1020	0.0	176.4
c102	1150.0	1090	5.5	1.8	1142	0.7	1211.3	1150	0.0	27.3	1150	1150	0.0	274.1
c103	1210.0	1150	5.2	2.5	1186	2.0	2329.6	1208	0.2	43.5	1210	1210	0.0	301.1
c104	1260.0	1220	3.3	3	1226	2.8	1493.9	1240	1.6	38.1	1246	1260	0.0	281.8
c105	1060.0	1030	2.9	1.8	1052	0.8	716.5	1060	0.0	18.6	1060	1060	0.0	333.2
c106	1080.0	1040	3.8	2.1	1058	2.1	556.8	1080	0.0	31.7	1080	1080	0.0	179.4
c107	1120.0	1100	1.8	2	1114	0.5	411.3	1120	0.0	23.4	1120	1120	0.0	209.3
c108	1130.0	1100	2.7	3.6	1112	1.6	820.1	1126	0.4	24	1126	1130	0.0	373.1
c109	1190.0	1180	0.8	2.5	1172	1.5	916	1182	0.7	27.5	1182	1190	0.0	227.6
c201	1810.0	1810	0.0	1.1	1810	0.0	1.2	1810	0.0	1.9	1810	1810	0.0	0.1
c202	1810.0	1810	0.0	1.1	1810	0.0	8.9	1810	0.0	0.9	1810	1810	0.0	0.1
c203	1810.0	1810	0.0	1	1810	0.0	43.9	1810	0.0	0.5	1810	1810	0.0	0.2
c204	1810.0	1810	0.0	1	1810	0.0	1.3	1810	0.0	0.2	1810	1810	0.0	0.1
c205	1810.0	1810	0.0	1	1810	0.0	0.3	1810	0.0	0.9	1810	1810	0.0	0.1
c206	1810.0	1810	0.0	1	1810	0.0	0.1	1810	0.0	0.2	1810	1810	0.0	0.1
c207	1810.0	1810	0.0	1	1800	0.6	5.8	1810	0.0	0.3	1810	1810	0.0	0.1
c208	1810.0	1810	0.0	0.8	1810	0.0	0.2	1810	0.0	0.2	1810	1810	0.0	0.1
r101	611.0	601	1.7	1.4	608	0.5	55.1	603.6	1.2	22.3	608	608	0.5	93.5
r102	843.0	807	4.5	1.7	825.6	2.1	1924.5	828	1.8	19.9	834	837	0.7	162.6
r103	928.0	878	5.7	2.2	902.2	2.9	2622.2	920.2	0.8	37	924	928	0.0	206.9
r104	969.0	941	3.0	3.8	944.2	2.6	2343.9	961	0.8	36.3	967	969	0.0	209.1
r105	778.0	735	5.9	2.9	766	1.6	838.5	753.8	3.2	19.8	757	778	0.0	142.4
r106	906.0	870	4.1	3.5	889.8	1.8	929.5	897.6	0.9	30.9	905	906	0.0	185.9
r107	950.0	927	2.5	3.3	932.4	1.9	2660.9	937.2	1.4	36.3	945	950	0.0	221.4
r108	994.0	982	1.2	3.2	976.2	1.8	2596.2	967	2.8	46.7	970	994	0.0	220
r109	885.0	866	2.2	2.1	876.6	1.0	1374.5	885	0.0	31.1	885	885	0.0	145.8
r110	915.0	870	5.2	2	900	1.7	926.3	889	2.9	24.3	890	915	0.0	168.6
r111	952.0	935	1.8	2	932.4	2.1	1896.4	948	0.4	34.2	949	952	0.0	203.5
r112	967.0	939	3.0	3.1	947.6	2.0	1662.6	948	2.0	28.8	957	967	0.0	251.8
r201	1458.0	1458	0.0	1.3	1458	0.0	376.4	1458	0.0	1	1458	1458	0.0	0.2
r202	1458.0	1458	0.0	1.1	1458	0.0	936.2	1458	0.0	0.5	1458	1458	0.0	0.2
r203	1458.0	1458	0.0	0.9	1458	0.0	73.9	1458	0.0	0.2	1458	1458	0.0	0.2
r204	1458.0	1458	0.0	0.6	1458	0.0	0	1458	0.0	0.1	1458	1458	0.0	0.2
r205	1458.0	1458	0.0	0.9	1458	0.0	1.3	1458	0.0	0.3	1458	1458	0.0	0.2
r206	1458.0	1458	0.0	0.9	1458	0.0	0	1458	0.0	0.2	1458	1458	0.0	0.1
r207	1458.0	1458	0.0	0.8	1458	0.0	0	1458	0.0	0.1	1458	1458	0.0	0.2
r208	1458.0	1458	0.0	0.5	1458	0.0	0	1458	0.0	0.1	1458	1458	0.0	0.2
r209	1458.0	1458	0.0	1	1458	0.0	0	1458	0.0	0.2	1458	1458	0.0	0.2
r210	1458.0	1458	0.0	0.9	1458	0.0	3.1	1458	0.0	0.2	1458	1458	0.0	0.2
r211	1458.0	1458	0.0	0.7	1458	0.0	0	1458	0.0	0.1	1458	1458	0.0	0.2
rc101	808.0	794	1.8	1.9	805.4	0.3	1324.3	802.6	0.7	22.5	810	808	0.0	119.7
rc102	902.0	881	2.4	2.3	899.4	0.3	2218.8	902	0.0	44.5	903	899	0.3	136.7
rc103	974.0	947	2.9	2	941.8	3.4	2005.2	950	2.5	37.6	954	974	0.0	170.2
rc104	1064.0	1019	4.4	1.7	1013.2	5.0	2139.3	1059	0.5	28.5	1059	1064	0.0	194.1
rc105	875.0	841	4.0	1.5	867.2	0.9	1052.3	861.8	1.5	20.2	862	875	0.0	127
rc106	909.0	874	4.0	2.5	901.4	0.8	2106.7	897.4	1.3	34.7	900	909	0.0	143
rc107	980.0	951	3.0	1.9	959.2	2.2	1763.1	977.2	0.3	26	980	980	0.0	155.4
rc108	1023.0	998	2.5	2	1000.2	2.3	2222.3	1023	0.0	27.5	1025	1020	0.3	173.2
rc201	1724.0	1724	0.0	2.1	1724	0.0	2327.4	1724	0.0	2	1724	1724	0.0	0.2
rc202	1724.0	1724	0.0	1.1	1724	0.0	1095.8	1724	0.0	0.7	1724	1724	0.0	0.2
rc203	1724.0	1724	0.0	0.9	1724	0.0	308.6	1724	0.0	0.4	1724	1724	0.0	0.2
rc204	1724.0	1724	0.0	0.8	1724	0.0	53.3	1724	0.0	0.3	1724	1724	0.0	0.2
rc205	1724.0	1724	0.0	2.1	1724	0.0	1313.7	1724	0.0	1.2	1724	1724	0.0	0.2
rc206	1724.0	1724	0.0	1	1723.6	0.0	2.8	1724	0.0	0.8	1724	1724	0.0	0.2
rc207	1724.0	1724	0.0	1	1722.4	0.1	72.1	1724	0.0	0.5	1724	1724	0.0	0.2
rc208	1724.0	1724	0.0	0.9	1724	0.0	0	1724	0.0	0.3	1724	1724	0.0	0.2
Average		1283.5	1.7	1.7	1290.5	0.9	907.0	1294.8	0.5	15.6	1296.2	1310.6	0.0	107.3

seconds required by the ILS heuristic, the average profit of five runs obtained by the ACS heuristic, the percentage gap between the best-known solution and the obtained solution, and the average computation time in seconds, the average profit obtained by the I3CH heuristic, the percentage gap between the best-known solution and the obtained solution, and the average computation time in seconds, respectively. Note that the GVNS, VNS, SSA, ILS, ACS and I3CH algorithms were not run on our computer, and the CPU times are taken from the original papers. The 23th, 24th, 25th, and 26th columns present the average profit obtained by

Table 8
Results for the test problems of Solomon (new data set).

Name	m	BKS	GVNS			GRASP-ELS			VNS			SSA		
			P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)
c101	10	1810.0	1754.0	3.2	27.0	1810.0	0.0	8.9	1809.0	0.1	21.7	1770.0	2.3	89.5
c102	10	1810.0	1794.0	0.9	23.2	1810.0	0.0	0.3	1810.0	0.0	19.8	1810.0	0.0	52.9
c103	10	1810.0	1810.0	0.0	5.7	1810.0	0.0	0.4	1810.0	0.0	18.8	1810.0	0.0	52.9
c104	10	1810.0	1810.0	0.0	1.6	1810.0	0.0	0.4	1810.0	0.0	16.8	1810.0	0.0	44.4
c105	10	1810.0	1810.0	0.0	1.6	1810.0	0.0	0.3	1810.0	0.0	22.7	1780.0	1.7	152.7
c106	10	1810.0	1806.0	0.2	9.1	1810.0	0.0	0.1	1808.0	0.1	23.5	1800.0	0.6	123.1
c107	10	1810.0	1810.0	0.0	1.1	1810.0	0.0	0.1	1810.0	0.0	21.9	1760.0	2.8	61.1
c108	10	1810.0	1810.0	0.0	0.7	1810.0	0.0	0.2	1810.0	0.0	19.6	1770.0	2.2	59.6
c109	10	1810.0	1810.0	0.0	0.1	1810.0	0.0	0.0	1810.0	0.0	16.1	1810.0	0.0	63.0
c201	4	1810.0	1810.0	0.0	0.1	1810.0	0.0	0.0	1810.0	0.0	92.5	1810.0	0.0	44.9
c202	4	1810.0	1810.0	0.0	1.8	1810.0	0.0	0.0	1810.0	0.0	114.8	1810.0	0.0	44.2
c203	4	1810.0	1810.0	0.0	1.5	1810.0	0.0	0.0	1810.0	0.0	84.7	1810.0	0.0	42.0
c204	4	1810.0	1810.0	0.0	0.5	1810.0	0.0	0.0	1810.0	0.0	107.0	1810.0	0.0	39.6
c205	4	1810.0	1810.0	0.0	0.1	1810.0	0.0	0.0	1810.0	0.0	106.6	1810.0	0.0	42.0
c206	4	1810.0	1810.0	0.0	0.1	1810.0	0.0	0.0	1810.0	0.0	109.3	1810.0	0.0	40.6
c207	4	1810.0	1810.0	0.0	0.1	1810.0	0.0	0.0	1810.0	0.0	115.0	1810.0	0.0	40.8
c208	4	1810.0	1810.0	0.0	0.1	1810.0	0.0	0.0	1810.0	0.0	108.4	1810.0	0.0	40.0
r101	19	1458.0	1432.2	26.4	23.4	1454.4	0.2	60.9	1455.8	0.2	9.2	1455.0	0.2	67.9
r102	17	1458.0	1441.2	25.6	20.7	1451.8	0.4	146.5	1451.1	0.5	9.6	1458.0	0.0	84.9
r103	13	1458.0	1446.6	25.1	23.5	1455.0	0.2	178.6	1453.3	0.3	16.9	1455.0	0.2	59.6
r104	9	1458.0	1418.2	27.6	55.0	1445.0	0.9	103.9	1443.5	1.0	28.7	1442.0	1.1	131.3
r105	14	1458.0	1441.6	25.6	32.3	1449.6	0.6	74.4	1450.1	0.5	16.9	1458.0	0.0	136.5
r106	12	1458.0	1437.6	25.9	25.3	1453.8	0.3	89.1	1451.9	0.4	20.2	1458.0	0.0	116.3
r107	10	1458.0	1435.0	26.1	46.0	1450.6	0.5	96.6	1449.7	0.6	26.9	1452.0	0.4	78.8
r108	9	1458.0	1441.8	25.5	56.4	1455.0	0.2	116.8	1453.2	0.3	26.6	1447.0	0.8	87.1
r109	11	1458.0	1433.4	26.3	42.8	1439.8	1.2	78.6	1448.3	0.7	23.1	1453.0	0.3	128.1
r110	10	1458.0	1433.4	26.3	50.1	1434.2	1.6	102.6	1450.5	0.5	26.4	1454.0	0.3	172.9
r111	10	1458.0	1430.2	26.6	40.0	1438.6	1.3	88.6	1449.2	0.6	25.5	1444.0	1.0	114.6
r112	9	1458.0	1434.4	26.2	58.7	1440.4	1.2	127.3	1451.9	0.4	27.9	1446.0	0.8	78.0
r201	4	1458.0	1458.0	24.1	1.4	1458.0	0.0	0.2	1458.0	0.0	294.6	1458.0	0.0	42.0
r202	3	1458.0	1450.2	24.8	19.8	1456.2	0.1	19.2	1453.7	0.3	519.7	1458.0	0.0	59.0
r203	3	1458.0	1458.0	24.1	2.8	1458.0	0.0	0.0	1458.0	0.0	257.3	1458.0	0.0	39.9
r204	2	1458.0	1452.6	24.6	6.6	1458.0	0.0	6.1	1455.3	0.2	785.2	1447.0	0.8	70.9
r205	3	1458.0	1458.0	24.1	0.4	1458.0	0.0	0.0	1458.0	0.0	296.3	1458.0	0.0	39.3
r206	3	1458.0	1458.0	24.1	0.7	1458.0	0.0	0.0	1458.0	0.0	240.3	1458.0	0.0	38.9
r207	2	1458.0	1449.8	24.8	12.4	1456.0	0.1	19.9	1453.6	0.3	723.2	1452.0	0.4	158.8
r208	2	1458.0	1458.0	24.1	0.9	1458.0	0.0	0.1	1458.0	0.0	448.7	1457.0	0.1	56.0
r209	3	1458.0	1458.0	24.1	0.2	1458.0	0.0	0.0	1458.0	0.0	272.0	1458.0	0.0	40.1
r210	3	1458.0	1458.0	24.1	0.9	1458.0	0.0	0.1	1458.0	0.0	289.9	1458.0	0.0	40.0
r211	2	1457.0	1452.6	24.6	14.6	1456.0	0.1	26.5	1457.0	0.0	957.5	1451.0	0.4	66.9
rc101	14	1724.0	1690.2	7.1	28.1	1705.2	1.1	65.4	1704.5	1.1	18.4	1712.0	0.7	79.9
rc102	12	1724.0	1685.0	7.4	30.1	1693.8	1.8	74.6	1696.0	1.6	22.0	1718.0	0.3	103.4
rc103	11	1724.0	1709.0	5.9	37.9	1716.4	0.4	129.3	1715.8	0.5	22.1	1724.0	0.0	82.6
rc104	10	1724.0	1718.0	5.4	43.6	1724.0	0.0	4.0	1722.7	0.1	21.5	1719.0	0.3	87.7
rc105	13	1724.0	1689.8	7.1	32.9	1694.4	1.7	90.4	1698.2	1.5	18.1	1716.0	0.5	87.0
rc106	11	1724.0	1690.6	7.1	40.5	1691.8	1.9	77.8	1694.8	1.7	24.5	1714.0	0.6	63.7
rc107	11	1724.0	1718.4	5.3	51.5	1724.0	0.0	66.4	1721.0	0.2	19.8	1722.0	0.1	51.4
rc108	10	1724.0	1713.0	5.7	51.1	1718.8	0.3	121.7	1721.8	0.1	21.7	1719.0	0.3	121.0
rc201	4	1724.0	1724.0	5.0	1.9	1724.0	0.0	0.0	1724.0	0.0	231.9	1724.0	0.0	41.7
rc202	3	1719.0	1702.8	6.3	10.3	1719.8	0.2	20.3	1711.6	0.4	491.2	1714.0	0.3	48.3
rc203	3	1724.0	1724.0	5.0	2.2	1724.0	0.0	0.7	1724.0	0.0	313.9	1724.0	0.0	39.1
rc204	3	1724.0	1724.0	5.0	0.1	1724.0	0.0	0.0	1724.0	0.0	222.8	1724.0	0.0	38.3
rc206	3	1724.0	1724.0	5.0	2.5	1724.0	0.0	0.7	1724.0	0.0	339.9	1724.0	0.0	46.0
rc207	3	1724.0	1724.0	5.0	2.2	1724.0	0.0	1.1	1724.0	0.0	295.1	1724.0	0.0	40.4
rc208	3	1724.0	1724.0	5.0	0.2	1724.0	0.0	0.0	1724.0	0.0	209.5	1724.0	0.0	39.1
Average			1628.9	12.2	17.2	1634.6	0.3	36.3	1635.1	0.3	157.0	1633.4	0.4	71.1

Name	m	BKS	ILS			ACS			ABC				I3CH		
			P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{max}	P_{avg}	Gap (%)	T (s)
c101	10	1810	1720.0	5.2	4.1	N/A	N/A	N/A	1786.0	1.3	59.1	1794.0	1810.0	0.0	1.3
c102	10	1810	1790.0	1.1	4.2	N/A	N/A	N/A	1810.0	0.0	14.3	1810.0	1810.0	0.0	3.6
c103	10	1810	1810.0	0.0	3.0	N/A	N/A	N/A	1810.0	0.0	6.4	1810.0	1810.0	0.0	0.7
c104	10	1810	1810.0	0.0	1.8	N/A	N/A	N/A	1810.0	0.0	1.9	1810.0	1810.0	0.0	0.6
c105	10	1810	1770.0	2.2	2.8	N/A	N/A	N/A	1782.0	1.5	58.8	1791.0	1810.0	0.0	2.8
c106	10	1810	1750.0	3.3	3.8	N/A	N/A	N/A	1786.0	1.3	53.2	1789.0	1810.0	0.0	178.9
c107	10	1810	1790.0	1.1	3.1	N/A	N/A	N/A	1784.0	1.4	53.2	1789.0	1810.0	0.0	9.1
c108	10	1810	1810.0	0.0	2.5	N/A	N/A	N/A	1792.0	1.0	44.2	1797.0	1810.0	0.0	230.6
c109	10	1810	1810.0	0.0	2.0	N/A	N/A	N/A	1810.0	0.0	24.1	1810.0	1810.0	0.0	0.7
c201	4	1810	1810.0	0.0	1.1	1810	0	1.2	1810.0	0.0	1.9	1810.0	1810.0	0.0	0.5
c202	4	1810	1810.0	0.0	1.1	1810	0	8.9	1810.0	0.0	0.9	1810.0	1810.0	0.0	0.6
c203	4	1810	1810.0	0.0	1.0	1810	0	43.9	1810.0	0.0	0.5	1810.0	1810.0	0.0	0.6
c204	4	1810	1810.0	0.0	1.0	1810	0	1.3	1810.0	0.0	0.2	1810.0	1810.0	0.0	0.7

Table 8 (*continued*)

Name	m	BKS	ILS			ACS			ABC				I3CH		
			P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	$P_{\max}$	P_{avg}	Gap (%)	T (s)
c205	4	1810	1810.0	0.0	1.0	1810	0	0.3	1810.0	0.0	0.9	1810.0	1810.0	0.0	0.5
c206	4	1810	1810.0	0.0	1.0	1810	0	0.1	1810.0	0.0	0.2	1810.0	1810.0	0.0	0.6
c207	4	1810	1810.0	0.0	1.0	1800	0.6	5.8	1810.0	0.0	0.3	1810.0	1810.0	0.0	0.6
c208	4	1810	1810.0	0.0	0.8	1810	0	0.2	1810.0	0.0	0.2	1810.0	1810.0	0.0	0.6
r101	19	1458	1441.0	1.2	2.5	N/A	N/A	N/A	1457.4	0.0	25.1	1458.0	1458.0	0.0	1.2
r102	17	1458	1450.0	0.5	3.1	N/A	N/A	N/A	1455.4	0.2	18.4	1458.0	1458.0	0.0	1.9
r103	13	1458	1450.0	0.5	2.0	N/A	N/A	N/A	1455.0	0.2	14.3	1458.0	1458.0	0.0	958.1
r104	9	1458	1402.0	3.8	2.3	N/A	N/A	N/A	1437.8	1.4	59.0	1440.0	1454.0	0.3	1014.3
r105	14	1458	1435.0	1.6	4.1	N/A	N/A	N/A	1458.0	0.0	43.2	1458.0	1458.0	0.0	44.2
r106	12	1458	1441.0	1.2	3.1	N/A	N/A	N/A	1458.0	0.0	32.4	1458.0	1458.0	0.0	84.0
r107	10	1458	1431.0	1.9	3.3	N/A	N/A	N/A	1452.0	0.4	45.8	1458.0	1458.0	0.0	1069.5
r108	9	1458	1430.0	1.9	2.7	N/A	N/A	N/A	1451.8	0.4	64.9	1455.0	1456.0	0.1	1543.5
r109	11	1458	1432.0	1.8	2.5	N/A	N/A	N/A	1449.4	0.6	64.3	1458.0	1458.0	0.0	474.2
r110	10	1458	1419.0	2.7	4.4	N/A	N/A	N/A	1449.0	0.6	54.8	1451.0	1454.0	0.3	1281.1
r111	10	1458	1410.0	3.3	3.0	N/A	N/A	N/A	1449.8	0.6	47.5	1452.0	1458.0	0.0	571.8
r112	9	1458	1418.0	2.7	2.4	N/A	N/A	N/A	1448.0	0.7	43.1	1448.0	1456.0	0.1	3488.4
r201	4	1458	1458.0	0.0	1.3	1458	0	376.4	1458.0	0.0	1.0	1458.0	1458.0	0.0	0.7
r202	3	1458	1443.0	1.0	2.7	1445.6	0.9	1966.2	1458.0	0.0	5.8	1458.0	1458.0	0.0	5.5
r203	3	1458	1458.0	0.0	1.6	1452.4	0.4	2564.3	1458.0	0.0	0.8	1458.0	1458.0	0.0	0.7
r204	2	1458	1440.0	1.2	3.4	1399	4	2713.5	1453.4	0.3	35.6	1458.0	1458.0	0.0	35.7
r205	3	1458	1458.0	0.0	1.1	1454.8	0.2	1830.7	1458.0	0.0	1.3	1458.0	1458.0	0.0	0.8
r206	3	1458	1458.0	0.0	1.1	1456.8	0.1	1328.8	1458.0	0.0	0.6	1458.0	1458.0	0.0	0.7
r207	2	1458	1428.0	2.1	1.7	1385	5	2624.1	1448.4	0.7	35.7	1454.0	1456.0	0.1	948.8
r208	2	1458	1458.0	0.0	1.6	1425.4	2.2	2828.9	1457.8	0.0	12.6	1458.0	1458.0	0.0	4.5
r209	3	1458	1458.0	0.0	1.1	1456.8	0.1	1163.7	1458.0	0.0	0.9	1458.0	1458.0	0.0	0.7
r210	3	1458	1458.0	0.0	1.2	1456.2	0.1	1428.1	1458.0	0.0	0.6	1458.0	1458.0	0.0	0.8
r211	2	1457	1422.0	2.4	1.9	1403	3.7	2726.1	1451.2	0.4	29.3	1456.0	1449.0	0.6	908.3
rc101	14	1724	1686.0	2.2	4.3	N/A	N/A	N/A	1713.6	0.6	52.9	1717.0	1724.0	0.0	54.2
rc102	12	1724	1659.0	3.8	2.9	N/A	N/A	N/A	1705.2	1.1	38.8	1713.0	1724.0	0.0	125.3
rc103	11	1724	1689.0	2.0	3.4	N/A	N/A	N/A	1721.0	0.2	40.9	1724.0	1724.0	0.0	7.1
rc104	10	1724	1719.0	0.3	3.2	N/A	N/A	N/A	1723.4	0.0	43.7	1724.0	1724.0	0.0	8.7
rc105	13	1724	1691.0	1.9	3.9	N/A	N/A	N/A	1716.0	0.5	47.6	1724.0	1724.0	0.0	31.2
rc106	11	1724	1665.0	3.4	4.8	N/A	N/A	N/A	1706.0	1.0	50.5	1723.0	1724.0	0.0	54.2
rc107	11	1724	1701.0	1.3	2.4	N/A	N/A	N/A	1724.0	0.0	46.0	1724.0	1724.0	0.0	12.8
rc108	10	1724	1698.0	1.5	5.6	N/A	N/A	N/A	1720.6	0.2	47.5	1724.0	1724.0	0.0	165.6
rc201	4	1724	1724.0	0.0	2.1	1724	0	2327.4	1724.0	0.0	2.0	1724.0	1724.0	0.0	0.8
rc202	3	1719	1686.0	1.9	1.7	1701.4	1	2230.7	1717.4	0.1	12.7	1719.0	1719.0	0.3	1516.0
rc203	3	1724	1724.0	0.0	2.9	1719.6	0.3	1213.2	1724.0	0.0	2.1	1724.0	1724.0	0.0	0.8
rc204	3	1724	1724.0	0.0	1.0	1722.2	0.1	1261.3	1724.0	0.0	1.1	1724.0	1724.0	0.0	0.8
rc206	3	1724	1708.0	0.9	1.3	1712.6	0.7	2185.2	1724.0	0.0	4.3	1724.0	1724.0	0.0	0.8
rc207	3	1724	1713.0	0.6	1.5	1714.8	0.5	2553.9	1724.0	0.0	1.9	1724.0	1724.0	0.0	0.8
rc208	3	1724	1724.0	0.0	1.1	1722.2	0.1	893.9	1724.0	0.0	1.1	1724.0	1724.0	0.0	0.9
Average			1620.9	1.1	2.4	1626.1	0.8	1318.4	1634.2	0.3	24.6	1636.4	1638.8	0.0	1636.4

Table 9
Results for the test problems of Cordeau et al. ($m = 1$).

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P	Gap (%)	T (s)
pr01	308	307.2	0.3	1.2	308.0	0.0	81.2	305.0	1.0	8.3	308.0	0.0	1.2
pr02	404	403.6	0.1	3.7	403.9	0.0	251.5	404.0	0.0	29.1	402.6	0.3	3.0
pr03	394	388.0	1.5	4.1	390.5	0.9	419.8	394.0	0.0	59.9	394.0	0.0	3.1
pr04	489	475.4	2.9	14.7	488.1	0.2	733.2	489.0	0.0	106.7	474.6	3.0	5.3
pr05	594	578.0	2.8	20.5	586.1	1.3	1663.2	589.0	0.8	281.7	581.0	2.2	9.1
pr06	590	584.2	1.0	29.0	588.2	0.3	1628.4	575.0	2.6	253.4	583.4	1.1	8.6
pr07	298	297.0	0.3	1.7	297.5	0.2	141.5	298.0	0.0	15.0	294.4	1.2	1.6
pr08	463	463.0	0.0	3.8	452.2	2.4	450.2	462.0	0.2	76.0	462.8	0.0	3.3
pr09	490	482.0	1.7	12.3	481.5	1.8	1079.1	482.0	1.7	102.3	477.8	2.6	7.0
pr10	588.4	564.4	4.3	32.7	575.5	2.2	1772.4	578.0	1.8	189.7	574.2	2.5	8.1
pr11	353	329.0	7.3	1.9	328.0	7.6	131.4	351.0	0.6	10.3	329.2	7.2	2.0
pr12	442	435.0	1.6	6.5	442.0	0.0	309.6	430.0	2.8	26.3	442.0	0.0	4.0
pr13	466	452.4	3.0	16.2	453.7	2.7	514.1	452.0	3.1	49.0	456.6	2.1	4.9
pr14	560.1	540.6	3.6	32.4	550.1	1.8	1163.8	540.0	3.7	134.3	541.2	3.5	8.8
pr15	707	656.6	7.7	29.4	663.2	6.6	1900.9	666.0	6.2	118.5	665.0	6.3	13.1
pr16	652.6	643.4	1.4	60.9	637.0	2.4	1854.1	616.0	5.9	558.0	652.6	0.0	16.2
pr17	362	354.6	2.1	5.4	358.3	1.0	140.5	362.0	0.0	37.6	356.2	1.6	2.4
pr18	539	530.8	1.5	10.4	519.4	3.8	760.9	539.0	0.0	61.8	517.6	4.1	5.0

(*continued on next page*)

Table 9 (*continued*)

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P	Gap (%)	T (s)
pr19	551.6	507.8	8.6	22.1	551.6	0.0	1242.2	531.0	3.9	152.9	537.8	2.6	8.7
pr20	656.6	655.0	0.2	57.0	647.8	1.4	2441.9	626.0	4.9	475.3	656.6	0.0	13.9
Average		482.4	2.6	18.3	486.1	1.8	934.0	484.5	2.0	137.3	485.4	2.0	6.5

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{max}	P_{avg}	Gap (%)	T (s)
pr01	308.0	304.0	1.3	0.5	308.0	0.0	256.2	307.0	0.3	5.8	308.0	305.0	1.0	20.8
pr02	404.0	385.0	4.9	0.6	403.8	0.0	1147.8	391.2	3.3	4.9	392.0	394.0	2.5	47.9
pr03	394.0	384.0	2.6	1.0	394.0	0.0	2024.3	394.0	0.0	12.6	394.0	394.0	0.0	72.9
pr04	489.0	447.0	9.4	1.9	482.6	1.3	1404.7	486.0	0.6	67.4	489.0	489.0	0.0	109.3
pr05	594.0	576.0	3.1	4.6	576.8	3.0	2075.7	586.8	1.2	91.4	589.0	594.0	0.0	185.4
pr06	590.0	538.0	9.7	2.5	564.6	4.5	2199.8	573.6	2.9	247.4	576.0	590.0	0.0	189.9
pr07	298.0	291.0	2.4	0.4	298.0	0.0	20.1	298.0	0.0	4.3	298.0	298.0	0.0	26.5
pr08	463.0	463.0	0.0	1.0	462.6	0.1	2476.0	462.0	0.2	15.0	463.0	454.0	2.0	77.4
pr09	490.0	461.0	6.3	1.4	481.8	1.7	2318.2	481.8	1.7	126.3	482.0	490.0	0.0	137.8
pr10	588.4	539.0	9.2	3.6	588.4	0.0	2343.5	570.4	3.2	210.3	574.0	568.0	3.6	222.2
pr11	353.0	330.0	7.0	0.3	327.8	7.7	1743.7	350.6	0.7	1.7	353.0	353.0	0.0	30.8
pr12	442.0	431.0	2.6	0.9	436.4	1.3	2017.6	432.0	2.3	23.5	432.0	433.0	2.1	59.8
pr13	466.0	450.0	3.6	1.9	441.0	5.7	2312.7	458.1	1.7	8.9	461.0	466.0	0.0	89.5
pr14	560.1	482.0	16.2	1.1	494.0	13.4	23.1	560.1	0.0	76.6	565.0	521.0	7.5	144.4
pr15	707.0	638.0	10.8	5.3	524.8	34.7	18.2	666.4	6.1	274.6	670.0	707.0	0.0	248.2
pr16	652.6	559.0	16.7	4.1	517.8	26.0	25.7	613.4	6.4	205.1	614.0	619.0	5.4	228.6
pr17	362.0	346.0	4.6	0.2	358.0	1.1	1330.9	359.0	0.8	6.6	359.0	360.0	0.6	34.7
pr18	539.0	479.0	12.5	0.8	488.8	10.3	1350.0	535.0	0.7	22.1	535.0	497.0	8.5	99.0
pr19	551.6	499.0	10.5	2.7	475.0	16.1	30.0	541.0	2.0	70.7	544.0	538.0	2.5	164.6
pr20	656.6	570.0	15.2	2.5	552.4	18.9	24.6	604.8	8.6	346.9	608.0	588.0	11.7	202.7
Average		458.6	7.4	1.9	458.8	7.3	1257.1	483.6	2.1	91.1	485.3	482.9	2.4	119.6

Table 10
Results for the test problems of Cordeau et al. ($m = 2$).

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P	Gap (%)	T (s)
pr01	502	495.0	1.4	3.6	487.3	3.0	80.1	502.0	0.0	20.2	502.0	0.0	3.8
pr02	714	706.4	1.1	12.3	696.9	2.5	216.0	712.0	0.3	37.3	701.0	1.9	10.1
pr03	741	718.4	3.1	19.2	715.3	3.6	356.0	741.0	0.0	217.7	732.0	1.2	14.3
pr04	917	903.8	1.5	33.3	906.4	1.2	578.5	905.0	1.3	245.0	905.0	1.3	21.4
pr05	1101	1073.6	2.6	94.3	1064.8	3.4	982.7	1053.0	4.6	249.8	1069.4	3.0	35.5
pr06	1070.2	1070.2	0.0	71.0	1004.4	6.6	932.9	1022.0	4.7	439.3	1058.8	1.1	29.1
pr07	566	557.8	1.5	6.0	550.1	2.9	130.0	566.0	0.0	20.5	560.2	1.0	6.5
pr08	826.2	824.4	0.2	20.0	799.6	3.3	344.7	822.0	0.5	75.6	826.2	0.0	13.2
pr09	883.4	883.4	0.0	49.7	855.7	3.2	608.5	854.0	3.4	120.7	858.6	2.9	27.0
pr10	1117	1109.0	0.7	81.5	1067.6	4.6	1018.9	1069.0	4.5	313.2	1072.8	4.1	33.7
pr11	566	539.4	4.9	3.7	537.8	5.2	91.8	566.0	0.0	17.5	542.8	4.3	5.2
pr12	768	750.8	2.3	28.0	747.0	2.8	233.8	759.0	1.2	39.0	755.8	1.6	14.8
pr13	832	827.0	0.6	41.3	801.3	3.8	334.4	825.0	0.8	107.0	814.4	2.2	19.9
pr14	999	999.0	0.0	117.3	990.6	0.8	725.0	922.0	8.4	111.7	955.6	4.5	34.7
pr15	1210.4	1210.4	0.0	126.6	1182.4	2.4	1019.1	1155.0	4.8	444.8	1179.6	2.6	50.1
pr16	1217.8	1217.8	0.0	209.6	1192.5	2.1	1192.0	1094.0	11.3	330.7	1148.6	6.0	67.0
pr17	652	626.0	4.2	9.6	646.1	0.9	128.0	643.0	1.4	20.2	642.2	1.5	8.2
pr18	937	914.4	2.5	39.5	904.0	3.7	416.7	929.0	0.9	88.2	915.8	2.3	19.2
pr19	1017	1004.4	1.3	120.3	970.5	4.8	819.5	1017.0	0.0	302.7	991.4	2.6	27.8
pr20	1224.4	1224.4	0.0	128.5	1173.1	4.4	1227.6	1154.0	6.1	554.5	1193.2	2.6	40.8
Average		882.8	1.4	60.8	864.7	3.3	571.8	865.5	2.7	187.8	871.3	2.3	24.1

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{max}	P_{avg}	Gap (%)	T (s)
pr01	502	471.0	6.6	0.5	502.0	0.0	602.2	502.0	0.0	4.1	502.0	502.0	0.0	51.8
pr02	714	660.0	8.2	1.2	705.2	1.2	1017.6	704.6	1.3	20.0	712.0	714.0	0.0	127.7
pr03	741	714.0	3.8	3.3	731.4	1.3	2083.0	738.4	0.4	75.7	742.0	731.0	1.4	175.6
pr04	917	863.0	6.3	4.1	883.6	3.8	2439.3	905.0	1.3	126.2	913.0	917.0	0.0	270.1
pr05	1101	1011.0	8.9	7.1	1021.8	7.8	2278.5	1045.0	5.4	220.7	1052.0	1101.0	0.0	410.0
pr06	1070.2	997.0	7.3	9.8	987.6	8.4	1724.1	1009.8	6.0	445.7	1013.0	1040.0	2.9	427.6
pr07	566	552.0	2.5	1.0	566.0	0.0	972.3	559.2	1.2	6.7	561.0	566.0	0.0	71.8
pr08	826.2	796.0	3.8	5.1	813.0	1.6	1730.0	806.6	2.4	74.6	810.0	824.0	0.3	184.1
pr09	883.4	867.0	1.9	5.2	862.0	2.5	3131.0	859.2	2.8	225.3	865.0	878.0	0.6	304.9
pr10	1117	1004.0	11.3	10.3	1064.6	4.9	2918.6	1067.4	4.6	297.9	1075.0	1117.0	0.0	447.0

Table 10 (*continued*)

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{max}	P_{avg}	Gap (%)	T (s)
pr11	566	542.0	4.4	0.7	544.8	3.9	880.3	558.2	1.4	2.1	561.0	559.0	1.3	71.3
pr12	768	727.0	5.6	1.3	763.4	0.6	2706.1	754.8	1.7	19.4	757.0	768.0	0.0	143.6
pr13	832	757.0	9.9	2.4	798.8	4.2	3240.7	831.0	0.1	39.5	831.0	832.0	0.0	238.6
pr14	999	925.0	8.0	8.1	943.6	5.9	2100.7	939.8	6.3	120.9	946.0	978.0	2.1	337.3
pr15	1210.4	1126.0	7.5	8.2	1100.0	10.0	3188.8	1145.4	5.7	356.1	1154.0	1205.0	0.4	479.1
pr16	1217.8	1110.0	9.7	11.0	1101.0	10.6	2834.0	1106.4	10.1	708.8	1110.0	1124.0	8.3	500.5
pr17	652	624.0	4.5	1.3	646.2	0.9	1378.0	652.0	0.0	16.3	652.0	639.0	2.0	117.0
pr18	937	877.0	6.8	2.9	888.4	5.5	1960.3	929.0	0.9	77.3	933.0	937.0	0.0	231.0
pr19	1017	955.0	6.5	5.5	940.8	8.1	2366.9	1004.2	1.3	277.5	1008.0	1003.0	1.4	386.1
pr20	1224.4	1056.0	15.9	10.7	1113.0	10.0	3192.3	1173.3	4.4	600.5	1184.0	1155.0	6.0	541.6
Average		831.7	7.0	5.0	848.9	4.6	2137.2	864.6	2.9	185.8	869.1	879.5	1.3	275.8

Table 11
Results for the test problems of Cordeau et al. ($m = 3$).

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P	Gap (%)	T (s)
pr01	622	608.4	2.2	3.0	613.5	1.4	79.2	614.0	1.3	18.7	619.6	0.4	7.0
pr02	939	937.8	0.1	18.6	919.2	2.2	206.0	939.0	0.0	42.7	929.0	1.1	19.1
pr03	1010	997.4	1.3	29.9	983.6	2.7	330.6	989.0	2.1	173.2	999.0	1.1	27.0
pr04	1286	1279.6	0.5	106.5	1261.9	1.9	592.9	1253.0	2.6	168.7	1260.4	2.0	42.7
pr05	1481	1464.0	1.2	154.9	1441.1	2.8	779.3	1431.0	3.5	226.6	1450.8	2.1	68.2
pr06	1501	1499.4	0.1	188.9	1411.4	6.3	844.7	1469.0	2.2	332.4	1478.4	1.5	66.5
pr07	742	736.6	0.7	11.8	730.2	1.6	136.1	742.0	0.0	24.9	736.4	0.8	12.4
pr08	1139	1122.6	1.5	44.8	1089.0	4.6	297.5	1131.0	0.7	157.1	1118.6	1.8	26.3
pr09	1272	1260.6	0.9	102.6	1190.8	6.8	542.3	1236.0	2.9	251.3	1220.2	4.2	49.7
pr10	1567	1563.4	0.2	198.0	1493.3	4.9	923.5	1477.0	6.1	574.5	1529.0	2.5	86.6
pr11	654	647.0	1.1	5.9	649.6	0.7	58.5	654.0	0.0	12.8	652.0	0.3	6.6
pr12	997	975.2	2.2	61.0	982.2	1.5	194.8	967.0	3.1	43.0	976.6	2.1	23.1
pr13	1145	1102.2	3.9	77.0	1081.3	5.9	309.5	1139.0	0.5	80.3	1092.8	4.8	24.5
pr14	1357.4	1357.4	0.0	130.9	1318.7	2.9	549.8	1289.0	5.3	166.3	1324.6	2.5	43.9
pr15	1654	1594.2	3.8	246.6	1597.1	3.6	885.0	1550.0	6.7	414.2	1563.0	5.8	72.7
pr16	1654.6	1654.6	0.0	413.5	1615.8	2.4	1035.3	1530.0	8.1	697.6	1631.0	1.4	92.2
pr17	841	822.4	2.3	15.2	824.3	2.0	114.5	838.0	0.4	38.0	823.2	2.2	10.5
pr18	1276	1270.8	0.4	63.5	1220.4	4.6	326.3	1262.0	1.1	164.1	1234.8	3.3	25.3
pr19	1403	1393.6	0.7	216.3	1339.1	4.8	630.9	1329.0	5.6	220.7	1330.4	5.5	54.2
pr20	1677.6	1677.6	0.0	277.4	1640.2	2.3	1070.1	1593.0	5.3	681.3	1646.6	1.9	76.5
Average		1198.2	1.2	118.3	1170.1	3.3	495.3	1171.6	2.9	224.4	1180.8	2.4	41.8

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{max}	P_{avg}	Gap (%)	T (s)
pr01	622	598.0	4.0	0.4	619.0	0.5	117.0	617.4	0.7	6.4	622.0	622.0	0.0	132.1
pr02	939	899.0	4.4	3.9	938.4	0.1	2333.1	930.9	0.9	42.6	934.0	936.0	0.3	260.6
pr03	1010	946.0	6.8	3.9	991.0	1.9	876.8	985.4	2.5	60.1	991.0	1010.0	0.0	301.5
pr04	1286	1195.0	7.6	9.0	1226.4	4.9	2909.0	1256.0	2.4	147.0	1258.0	1286.0	0.0	442.7
pr05	1481	1356.0	9.2	12.5	1386.6	6.8	2746.7	1429.8	3.6	400.2	1438.0	1481.0	0.0	650.3
pr06	1501	1376.0	9.1	19.2	1359.0	10.4	2719.7	1449.0	3.6	619.1	1462.0	1501.0	0.0	651.2
pr07	742	713.0	4.1	1.0	738.4	0.5	1330.6	738.2	0.5	7.7	741.0	738.0	0.5	260.4
pr08	1139	1082.0	5.3	4.3	1115.0	2.2	2544.3	1130.2	0.8	42.5	1132.0	1139.0	0.0	307.0
pr09	1272	1144.0	11.2	10.3	1210.0	5.1	3187.7	1208.2	5.3	305.3	1215.0	1272.0	0.0	503.1
pr10	1567	1473.0	6.4	27.9	1457.2	7.5	2873.1	1460.6	7.3	658.4	1465.0	1567.0	0.0	731.1
pr11	654	632.0	3.5	0.5	649.0	0.8	237.3	652.2	0.3	7.1	654.0	654.0	0.0	151.7
pr12	997	902.0	10.5	1.8	970.6	2.7	2883.7	966.0	3.2	26.3	973.0	997.0	0.0	294.3
pr13	1145	1046.0	9.5	8.2	1088.8	5.2	2515.5	1111.0	3.1	62.7	1122.0	1145.0	0.0	378.9
pr14	1357.4	1197.0	13.4	8.3	1258.6	7.8	2084.7	1279.4	6.1	140.9	1288.0	1315.0	3.2	533.7
pr15	1654	1488.0	11.2	14.6	1486.8	11.2	2671.8	1588.4	4.1	443.0	1592.0	1654.0	0.0	708.1
pr16	1654.6	1478.0	11.9	28.2	1481.8	11.7	2284.5	1521.2	8.8	700.1	1534.0	1609.0	2.8	818.1
pr17	841	808.0	4.1	0.9	821.0	2.4	832.0	826.6	1.7	19.6	832.0	841.0	0.0	184.3
pr18	1276	1165.0	9.5	6.0	1209.0	5.5	2854.3	1241.2	2.8	88.6	1253.0	1276.0	0.0	386.6
pr19	1403	1238.0	13.3	10.2	1299.4	8.0	3094.8	1340.2	4.7	310.5	1353.0	1403.0	0.0	604.1
pr20	1677.6	1514.0	10.8	18.2	1491.2	12.5	2823.0	1587.8	5.7	677.6	1601.0	1658.0	1.2	909.7
Average		1112.5	8.3	9.5	1139.9	5.4	2196.0	1166.0	3.4	238.3	1173.0	1205.2	0.4	460.5

the ABC heuristic over the five runs, the gap between the best-known solution and the obtained solution in percentage, the average computation time in seconds when the best objective value is found for each run by the ABC heuristic, and the maximum profit obtained by the ABC algorithm over five runs P_{max}. In the last row of these tables, the average of each column is calculated.

Tables 4–7 show that the average gaps for ABC ranges from 0.5% to 0.7% for Solomon's (1987) test problems with $m = 1, 2, 3$, and 4. The average computation times range from 12.6 s to 21.5 s. Thus, we can say that the ABC algorithm obtains solutions that are very close to the best-known solutions within reasonable computational times. When the performances of the five algorithms are

Table 12
Results for the test problems of Cordeau et al. ($m = 4$).

Name	BKS	GVNS			VNS			SSA			GRASP-ELS		
		P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)	P	Gap (%)	T (s)
pr01	657	655.8	0.2	3.6	656.4	0.1	50.1	657.0	0.0	15.5	657.0	0.0	1.2
pr02	1073	1064.2	1.4	25.6	1057.8	1.4	196.3	1069.0	0.4	44.0	1062.8	1.0	21.0
pr03	1232	1209.6	1.0	57.2	1195.2	3.1	271.0	1201.0	2.6	156.1	1200.0	2.7	28.0
pr04	1585	1525.8	2.0	137.2	1502.1	5.5	436.0	1535.0	3.3	393.2	1524.6	4.0	49.7
pr05	1838	1811.4	1.2	208.4	1775.9	3.5	653.7	1759.0	4.5	321.0	1774.8	3.6	77.2
pr06	1840.4	1840.4	0.7	309.9	1798.1	2.4	782.7	1839.0	0.1	721.0	1820.8	1.1	85.1
pr07	872	862.6	1.1	17.9	849.0	2.7	97.8	871.0	0.1	31.6	858.0	1.6	11.3
pr08	1377	1358.2	1.7	67.3	1336.3	3.0	268.5	1358.0	1.4	118.8	1349.8	2.0	30.2
pr09	1604	1595.4	0.2	171.1	1492.4	7.5	475.3	1565.0	2.5	252.5	1562.4	2.7	59.2
pr10	1943	1903.6	1.5	275.1	1855.0	4.7	800.3	1853.0	4.9	502.0	1866.4	4.1	94.6
pr11	657	657.0	0.0	5.0	657.0	0.0	28.0	657.0	0.0	13.5	657.0	0.0	0.1
pr12	1130.1	1110.2	1.9	36.7	1117.6	1.1	143.7	1116.0	1.3	134.2	1121.0	0.8	26.7
pr13	1386	1324.8	1.6	76.6	1302.6	6.4	250.6	1355.0	2.3	112.6	1314.8	5.4	41.3
pr14	1651	1636.6	1.4	243.6	1612.9	2.4	423.5	1586.0	4.1	192.5	1623.4	1.7	72.2
pr15	2065	1933.4	1.3	428.3	1934.0	6.8	690.0	1929.0	7.1	518.6	1917.6	7.7	111.9
pr16	2017	2000.0	1.0	632.0	1968.9	2.4	824.5	1921.0	5.0	769.7	1976.6	2.0	149.6
pr17	934	914.6	2.0	14.0	924.7	1.0	78.9	926.0	0.9	30.5	927.2	0.7	15.2
pr18	1539	1505.4	1.0	93.0	1485.5	3.6	255.1	1455.0	5.8	104.8	1490.6	3.2	38.1
pr19	1750	1688.2	0.4	334.1	1659.8	5.4	523.3	1695.0	3.2	277.6	1632.0	7.2	77.0
pr20	2062	2012.0	1.2	463.1	1988.3	3.7	862.4	1901.0	8.5	685.8	1993.8	3.4	121.2
Average		1430.5	1.1	180.0	1408.5	3.3	405.6	1412.4	2.9	269.8	1416.5	2.7	55.5

Name	BKS	ILS			ACS			ABC				I3CH		
		P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{max}	P_{avg}	Gap (%)	T (s)
pr01	657	644.0	2.0	0.2	655.2	0.3	15.4	655.8	0.2	3.5	657.0	657.0	0.0	0.1
pr02	1073	1014.0	5.8	2.4	1067.8	0.5	2165.0	1053.6	1.8	38.0	1058.0	1073.0	0.0	380.6
pr03	1232	1162.0	6.0	10.5	1204.6	2.3	2074.0	1198.4	2.8	74.7	1199.0	1232.0	0.0	436.6
pr04	1585	1452.0	9.2	11.6	1506.0	5.2	2986.0	1525.0	3.9	199.9	1529.0	1585.0	0.0	603.6
pr05	1838	1665.0	10.4	19.6	1722.2	6.7	2818.5	1755.4	4.7	330.7	1761.0	1838.0	0.0	902.9
pr06	1840.4	1696.0	8.5	35.4	1723.5	6.8	2935.8	1763.6	4.4	678.3	1775.0	1835.0	0.3	939.6
pr07	872	840.0	3.8	1.6	865.4	0.8	1940.2	855.2	2.0	18.5	859.0	872.0	0.0	228.9
pr08	1377	1267.0	8.7	6.9	1351.6	1.9	2804.3	1336.8	3.0	87.1	1349.0	1377.0	0.0	429.9
pr09	1604	1460.0	9.9	13.8	1554.2	3.2	3371.9	1532.8	4.6	176.4	1543.0	1604.0	0.0	698.5
pr10	1943	1782.0	9.0	38.7	1819.8	6.8	3365.9	1842.0	5.5	725.7	1848.0	1943.0	0.0	1044.4
pr11	657	654.0	0.5	0.2	657.0	0.0	0.1	657.0	0.0	0.4	657.0	657.0	0.0	0.1
pr12	1130.1	1041.0	8.6	1.9	1114.0	1.4	2486.7	1130.1	0.0	38.7	1132.0	1120.0	0.9	477.1
pr13	1386	1263.0	9.7	6.6	1316.4	5.3	2926.3	1332.4	4.0	70.2	1342.0	1386.0	0.0	672.0
pr14	1651	1528.0	8.0	16.6	1534.4	7.6	2730.2	1569.2	5.2	211.5	1581.0	1651.0	0.0	783.2
pr15	2065	1818.0	13.6	19.5	1822.8	13.3	2631.3	1937.6	6.6	450.5	1940.0	2065.0	0.0	1161.7
pr16	2017	1889.0	6.8	35.9	1867.2	8.0	2614.7	1924.0	4.8	685.0	1928.0	2017.0	0.0	1183.8
pr17	934	889.0	5.1	1.9	923.8	1.1	2749.7	926.6	0.8	21.3	932.0	934.0	0.0	332.8
pr18	1539	1352.0	13.8	5.7	1464.8	5.1	3209.1	1462.8	5.2	86.5	1474.0	1539.0	0.0	559.5
pr19	1750	1560.0	12.2	22.2	1579.4	10.8	3293.9	1670.8	4.7	285.4	1680.0	1750.0	0.0	919.4
pr20	2062	1846.0	11.7	26.9	1821.4	13.2	3193.0	1958.8	5.3	826.3	1966.0	2062.0	0.0	1196.6
Average		1341.1	8.2	13.9	1378.6	5.0	2515.6	1404.4	3.5	250.4	1410.5	1459.9	0.1	647.6

Table 13
Results for the test problems of Cordeau et al. (new data set).

Name	m	BKS	GVNS			GRASP-ELS			VNS			SSA		
			P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P	Gap (%)	T (s)
pr01	3	657	608.4	7.4	3.1	619	5.8	10.1	613.5	6.6	79.2	614.0	6.5	18.7
pr02	6	1220	1198.8	1.7	27.9	1202.2	1.5	81.9	1195.3	2.0	85.0	1205.0	1.2	48.0
pr03	9	1788	1760.8	1.5	50.2	1771	1	171	1759.0	1.6	86.0	1758.0	1.7	124.2
pr04	12	2477	2467.4	0.4	114	2475.6	0.1	341.2	2466.4	0.4	81.2	2461.0	0.6	393.3
pr05	15	3351	3351	0	43.2	3351	0	3.5	3349.5	0.0	71.7	3351.0	0.0	1109.4
pr06	18	3671	3670.6	0	196.4	3671	0	31.5	3670.1	0.0	83.5	3661.0	0.3	1646.7
pr07	5	948	935	1.4	13.8	938.8	1	35.4	928.3	2.1	69.5	948.0	0.0	27.2
pr08	10	2006	2004.6	0.1	46.9	2006	0	3.8	2005.0	0.0	63.2	2006.0	0.0	128.5
pr09	15	2736	2736	0	8.5	2736	0	33.9	2735.6	0.0	54.6	2736.0	0.0	794.8
pr10	20	3850	3850	0	9.4	3850	0	2.5	3850.0	0.0	52.4	3847.0	0.1	1369.2
Average			2258.3	1.3	51.3	2262.1	0.9	71.5	2257.3	1.3	72.6	2258.7	1.0	566.0

Name	m	BKS	ILS			ACS			ABC				I3CH		
			P	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{avg}	Gap (%)	T (s)	P_{max}	P_{avg}	Gap (%)	T (s)
pr01	3	657	598	9	0.4	619	5.8	117	617.4	6	6.4	623	619.0	5.8	146.7
pr02	6	1220	1180	3.3	4.5	N/A	N/A	N/A	1203.8	1.3	32.3	1215	1207.0	1.1	669.7

Table 13 (*continued*)

Name	*m*	BKS	ILS			ACS			ABC				I3CH		
			P	Gap (%)	*T* (s)	P_{avg}	Gap (%)	*T* (s)	P_{avg}	Gap (%)	*T* (s)	P_{max}	P_{avg}	Gap (%)	*T* (s)
pr03	9	1788	1738	2.8	11.3	N/A	N/A	N/A	1764.6	1.3	138.6	1769	1781.0	0.4	1383.7
pr04	12	2477	2428	2	45.4	N/A	N/A	N/A	2474.2	0.1	266.7	2477	2477.0	0.0	641.9
pr05	15	3351	3297	1.6	37.3	N/A	N/A	N/A	3351	0	331.1	3351	3351.0	0.0	19.3
pr06	18	3671	3650	0.6	106.1	N/A	N/A	N/A	3670.6	0	752	3671	3671.0	0.0	30.0
pr07	5	948	909	4.1	1.5	N/A	N/A	N/A	931.4	1.8	14.9	936	943.0	0.5	299.7
pr08	10	2006	1984	1.1	12	N/A	N/A	N/A	2006	0	37.7	2006	2006.0	0.0	55.9
pr09	15	2736	2729	0.3	33	N/A	N/A	N/A	2736	0	104.8	2736	2736.0	0.0	10.8
pr10	20	3850	3850	0	52.3	N/A	N/A	N/A	3850	0	126.8	3850	3850.0	0.0	9.1
Average			2236.3	2.5	30.4	619	5.8	117	2260.5	1.1	181.1	2263.4	2264.1	0.8	326.7

Table 14
Average gaps for the GVNS, VNS, SSA, ILS, GRASP-ELS, ACS, ABC and I3CH (%).

Instance	GVNS		VNS		SSA		GRASP-ELS		ILS		ACS		ABC		I3CH	
	Gap (%)	*T* (s)	Gap (%)	*T* (s)	Gap (%)	*T* (s)	Gap (%)	*T* (s)	Gap (%)	*T* (s)	Gap (%)	*T* (s)	Gap (%)	*T* (s)	Gap (%)	*T* (s)
m = 1																
c100	1.3	166.5	0.1	98.4	0.0	21.1	0.0	22.6	1.1	0.3	0.0	6.4	0.1	2.6	0.0	25.2
r100	2.8	29.4	0.0	89.1	0.1	23.3	0.2	3.5	2.0	0.2	0.2	383.4	1.0	3.5	0.6	29.4
rc100	3.7	9.8	0.0	65.2	0.0	22.2	0.4	2.0	3.1	0.2	0.0	143.2	0.1	7.9	1.8	26.1
c200	1.0	192.4	0.1	560.2	0.0	37.5	0.5	32.2	2.2	1.7	0.1	560.2	0.4	11.1	0.3	84.4
r200	3.1	33.8	0.7	1065.8	1.0	45.8	1.3	11.2	2.6	1.7	0.7	1065.8	1.4	26.6	0.7	183.6
rc200	3.8	15.4	0.7	709.1	0.3	45.0	1.3	7.1	2.8	1.3	0.6	766.5	0.9	21.6	3.1	102.8
pr01–10	1.5	12.4	0.9	822.1	0.8	112.2	1.3	5.0	4.9	1.8	1.1	1626.6	1.3	78.5	0.9	109.0
pr11–20	3.7	24.2	2.7	1045.9	3.1	162.4	2.7	7.9	10.0	2.0	13.5	887.7	2.9	103.7	3.8	130.2
Average	2.6	60.5	0.7	557.0	0.7	58.7	1.0	11.4	3.6	1.1	2.0	680.0	1.0	31.9	1.4	86.3
m = 2																
c100	0.7	139.5	0.3	88.0	0.0	26.4	0.1	70.9	1.0	1.1	0.3	818.0	0.1	8.5	0.0	87.0
r100	1.7	60.3	1.3	63.5	0.1	36.6	1.6	8.0	2.3	0.9	0.6	1559.4	0.6	14.8	0.4	64.9
rc100	2.8	20.3	1.4	55.2	0.1	40.5	2.3	4.7	2.4	0.7	1.1	1375.8	0.2	15.4	0.8	59.8
c200	0.4	33.8	0.6	545.6	1.0	53.7	0.0	29.3	2.4	3.5	1.6	1398.1	0.8	19.2	0.5	401.2
r200	1.1	14.7	0.5	1015.1	0.4	91.4	0.4	17.6	2.6	2.3	3.6	2735.2	0.5	35.3	0.0	546.2
rc200	2.0	16.5	1.4	655.5	0.7	70.4	0.7	14.8	3.5	1.9	3.0	2033.5	1.2	32.2	0.3	374.9
pr01–10	1.2	39.1	3.4	524.8	1.9	173.9	1.6	19.5	6.1	4.8	3.1	1889.7	2.5	149.7	0.5	247.1
pr11–20	1.6	82.4	3.1	618.8	3.5	201.6	3.0	28.8	7.9	5.2	6.0	2384.8	3.2	221.8	2.2	304.6
Average	1.4	50.8	1.5	445.8	0.9	86.8	1.2	24.2	3.5	2.5	2.4	1774.3	1.1	62.1	0.6	260.7
m = 3																
c100	1.0	165.0	0.7	85.5	0.3	35.3	0.6	86.7	2.6	1.5	0.8	1043.2	0.5	16.3	0.1	190.2
r100	2.3	73.9	1.5	61.9	0.4	56.1	1.6	13.9	1.8	1.7	1.5	1668.9	0.5	25.1	0.2	122.9
rc100	2.3	33.7	1.3	60.6	0.6	42.8	2.8	8.7	3.2	1.1	2.1	1476.8	0.8	25.8	0.2	102.9
c200	0.2	7.7	0.0	241.6	1.3	59.7	0.0	0.1	2.0	2.2	1.7	1320.4	0.9	26.0	0.0	12.3
r200	0.2	7.0	0.1	321.7	0.1	41.9	0.0	2.5	0.3	1.4	0.2	1171.6	0.0	4.6	0.0	1.8
rc200	0.7	13.8	0.5	335.7	0.3	56.8	0.6	8.7	1.9	1.6	1.3	1546.5	0.3	13.9	0.1	104.8
pr01–10	0.9	85.9	3.5	473.2	2.1	197.0	1.8	40.6	6.8	9.2	4.0	2163.8	2.8	228.9	0.1	424.0
pr11–20	1.4	150.7	3.1	517.5	3.6	251.8	3.0	43.0	9.8	9.7	6.8	2228.2	4.0	247.6	0.7	497.0
Average	1.1	67.2	1.3	262.2	1.1	92.7	1.3	25.5	3.5	3.5	2.3	1577.4	1.2	73.5	0.2	182.0
m = 4																
c100	1.6	133.2	1.2	81.9	0.5	49.5	0.9	84.6	3.1	2.6	1.4	1056.1	0.3	27.5	0.0	261.8
r100	2.3	84.7	1.5	61.2	0.7	58.4	1.6	24.2	3.4	2.6	1.8	1652.6	1.5	30.6	0.1	192.5
rc100	1.7	36.9	2.2	58.5	0.2	68.1	2.2	13.4	3.1	2.0	1.9	1854.0	0.8	30.2	0.1	157.1
c200	0.0	0.6	0.0	104.8	0.0	41.8	0.0	0.0	0.0	1.0	0.1	7.7	0.0	0.6	0.0	0.1
r200	0.0	0.3	0.0	150.7	0.0	39.7	0.0	0.0	0.0	0.9	0.0	126.4	0.0	0.3	0.0	0.2
rc200	0.3	10.7	0.3	143.0	0.0	46.2	0.4	3.4	0.6	1.4	0.5	915.9	0.0	6.0	0.0	36.7
pr01–10	1.1	127.3	3.4	403.2	2.0	255.6	2.3	45.8	7.3	14.1	3.4	2447.7	3.3	233.3	0.0	566.5
pr11–20	1.2	232.6	3.3	408.0	3.8	284.0	3.2	65.3	9.0	13.7	6.6	2583.5	3.7	267.6	0.1	728.6
Average	1.0	78.3	1.5	176.4	0.9	105.4	1.3	29.6	3.3	4.8	2.0	1330.5	1.2	74.5	0.0	242.9

compared, the GVNS, VNS, SSA, GRASP-ELS, ILS, ACS, I3CH and ABC heuristics are dominant in 1, 3, 17, 4, 0, 1, 54 and 7 problems, respectively, out of the 224 problems in the four data sets. The performance of the GVNS, ILS, and ACS algorithms are worse than those of the other algorithms according to these results. The solution quality of the tree algorithms VNS, SSA, and ABC are quite similar, but the I3CH heuristic outperforms the others.

As shown in Table 8, for problems in Solomon's new data set for which optimal solutions are known, the average gaps of all seven algorithms are quite low. Thus, all of the algorithms perform similarly and yield solutions that are very similar to the optimal solutions for these problems.

Tables 9–12 show that the average gap for ABC ranges from 2.1% to 3.5% for the test problems of Cordeau et al. (1997), with m = 1, 2, 3, and 4. This time, the average computation times range from 91.1 s to 250.4 s. Although the performance of the ABC algorithm is reduced for these test problems, we can say that the ABC algorithm still obtains solutions that are close to the best-known solutions within reasonable computation times. When the six algorithm performances are compared, we can see that the GVNS,

VNS, SSA, GRASP-ELS, ILS, ACS, ABC and I3CH heuristics are dominant in 10, 0, 8, 1, 0, 1, 3 and 43 problems, respectively, out of the 80 problems in these data sets. This time, the performance of the VNS, GRASP-ELS, ILS and ACS algorithms are worse than those of the other algorithms. However, the I3CH algorithm clearly outperforms the other heuristics.

As shown in Table 13, for problems in Cordeau et al.'s new data set, the average gaps of all six algorithms (excluding ACS, which was tested only for one instance) are quite similar, ranging from 0.9% to 2.5%. Thus, all of the algorithms perform similarly and obtain near optimum solutions for these problems.

For an overall discussion of the findings, we present Table 14. As seen in this table, the ABC algorithm is comparable to the other heuristics; namely, GVNS, VNS, SSA, GRASP-ELS, ILS, ACS, and I3CH. The overall average gaps for GVNS, VNS, SSA, GRASP-ELS, ILS, ACS, ABC, and I3CH are 1.5%, 1.2%, 0.9%, 1.2%, 3.5%, 2.2%, 1.1%, and 0.6%, respectively. With respect to efficiency, we can say that all of the algorithms find solutions within reasonable computational times. The overall performances of the GVNS, VNS, SSA, GRASP-ELS, ILS, ACS, ABC, and I3CH methods averaged 64.2, 360.3, 85.9, 22.7, 3.0, 1340.5, 60.5 and 193.0 s, respectively.

6. Conclusions

This study focuses on solving a highly constrained NP-hard combinatorial optimization problem, namely, the team orienteering problem with time windows (TOPTW), using a comparatively recently introduced technique called the artificial bee colony (ABC) algorithm, which simulates the foraging behavior of honey-bee swarms. This study contributes to the literature by presenting a use of the ABC method for a difficult discrete optimization problem. The ABC algorithm was initially introduced for numerical function optimization, and the number of studies that address discrete optimization with the ABC algorithm is comparatively low. Additionally, this study introduces a new food source acceptance criterion and a new scout bee search behavior, both of which significantly contribute to the solution quality. The results show that the modified ABC heuristic of this study is capable of producing high-quality TOPTW solutions. The proposed modifications may also be applied to any ABC approach to produce high-quality solutions for similar problems.

References

Archetti, C., Hertz, A., & Speranza, M. G. (2007). Metaheuristics for the team orienteering problem. *Journal of Heuristics, 13*, 49–76.

Bouly, H., Dang, D. C., & Moukrim, A. (2010). A memetic algorithm for the team orienteering problem. *4OR, 8*, 49–70.

Butt, S. E., & Cavalier, T. M. (1994). A heuristic for the multiple tour maximum collection problem. *Computers and Operations Research* (21), 101–111.

Chao, I., Golden, B., & Wasil, E. (1996). The team orienteering problem. *European Journal of Operational Research, 88*, 464–474.

Cordeau, J. F., Gendreau, M., & Laporte, G. (1997). A tabu search heuristic for periodic and multi-depot vehicle routing problems. *Networks, 30*, 105–119.

Dang, D. C., Guibadj, R. N., & Moukrim, A. (2013). An effective pso-inspired algorithm for the team orienteering problem. *European Journal of Operational Research, 229*(2), 332–344.

Feillet, D., Dejax, P., & Gendreau, M. (2005). Traveling salesman problems with profits. *Transportation Science, 39*, 188–205.

Hu, Q., & Lim, A. (2014). An iterative three-component heuristic for the team orienteering problem with time windows. *European Journal of Operational Research, 232*(2), 276–286.

Kantor, M. G., & Rosenwein, M. B. (1992). The orienteering problem with time windows. *Journal of the Operational Research Society, 43*(6), 629–635.

Karaboga, D. (2005) *An idea based on honey bee swarm for numerical optimization: Technical Report-TR06*. Erciyes University, Engineering Faculty, Computer Engineering Department.

Karaboga, D., & Basturk, B. (2007). A powerful and efficient algorithm for numerical function optimization: Artificial bee colony (ABC) algorithm. *Journal of Global Optimization, 39*, 459–471.

Ke, L., Archetti, C., & Feng, Z. (2008). Ants can solve the team orienteering problem. *Computers and Industrial Engineering, 54*, 648–665.

Labadie, N., Mansini, R., Melechovský, J., & Calvo, R. W. (2012). The team orienteering problem with time windows: An LP-based granular variable neighborhood search. *European Journal of Operational Research, 220*, 15–27.

Labadie, N., Melechovský, J., & Calvo, R. W. (2011). Hybridized evolutionary local search algorithm for the team orienteering problem with time windows. *Journal of Heuristics, 17*, 729–753.

Laporte, G., & Martello, S. (1990). The selective travelling salesman problem. *Discrete Applied Mathematics, 26*, 193–207.

Leuven, K. U. (2012). http://www.mech.kuleuven.be/en/cib/op.

Lin, S., & Yu, V. F. (2012). A simulated annealing heuristic for the team orienteering problem with time windows. *European Journal of Operational Research, 217*, 94–107.

Montemanni, R., & Gambardella, L. M. (2009). Ant colony system for team orienteering problems with time windows. *Foundations of Computing and Decision Sciences, 34*, 287–306.

Righini, G., & Salani, M. (2009). Decremental state space relaxation strategies and initialization heuristics for solving the orienteering problem with time windows with dynamic programming. *Computers and Operations Research, 36*, 1191–1203.

Singh, A. (2009). An artificial bee colony algorithm for the leaf-constrained minimum spanning tree problem. *Applied Soft Computing, 9*, 625–631.

Solomon, M. M. (1987). Algorithms for the vehicle routing and scheduling problems with time window constraints. *Operations Research, 35*, 254–265.

Souffriau, W., Vansteenwegen, P., Berghe, G. V., & Van Oudheusden, D. (2010). A path relinking approach for the team orienteering problem. *Computers and Operations Research, 37*, 1853–1859.

Szeto, W. Y., Yongzhong, W., & Ho, S. C. (2011). An artificial bee colony algorithm for the capacitated vehicle routing problem. *European Journal of Operational Research, 215*, 126–135.

Tang, H., & Miller-Hooks, E. (2005). A TABU search heuristic for the team orienteering problem. *Computers and Operations Research, 32*, 1379–1407.

Teodorovic, D., Lucic, P., Markovic, G., & Orco, M. D. (2006). Bee colony optimization: Principles and applications. In *Neural network applications in electrical engineering (NEUREL 2006)* (pp. 151–156). Belgrade.

Tricoire, F., Romauch, M., Doerner, K. F., & Hartl, R. F. (2010). Heuristics for the multiperiod orienteering problem with multiple time windows. *Computers and Operations Research, 37*, 351–367.

Tsiligirides, T. (1984). Heuristic methods applied to orienteering. *Journal of the Operational Research Society, 35*(9), 797–809.

Vansteenwegen, P., Souffriau, W., Berghe, G. V., & Van Oudheusden, D. (2009a). A guided local search metaheuristic for the team orienteering problem. *European Journal of Operational Research, 196*, 118–127.

Vansteenwegen, P., Souffriau, W., Berghe, G. V., & Van Oudheusden, D. (2009b). Iterated local search for the team orienteering problem with time windows. *Computers and Operations Research, 36*, 3281–3290.

Vansteenwegen, P., Souffriau, W., & Van Oudheusden, D. (2011). The orienteering problem: A survey. *European Journal of Operational Research, 209*(1), 1–10.